REPORT ON THE WATER CONSERVATION KNOWLEDGE-SHARING PLATFORM

1. Objective

The objective of our project is to create a centralized, accessible knowledge-sharing platform to promote water conservation techniques. By addressing the lack of awareness and accessibility, the platform aims to encourage sustainable practices, reduce water wastage, and contribute to mitigating water scarcity challenges globally. The project seeks to empower users with tools, techniques, and localized insights for water conservation, particularly in agriculture.

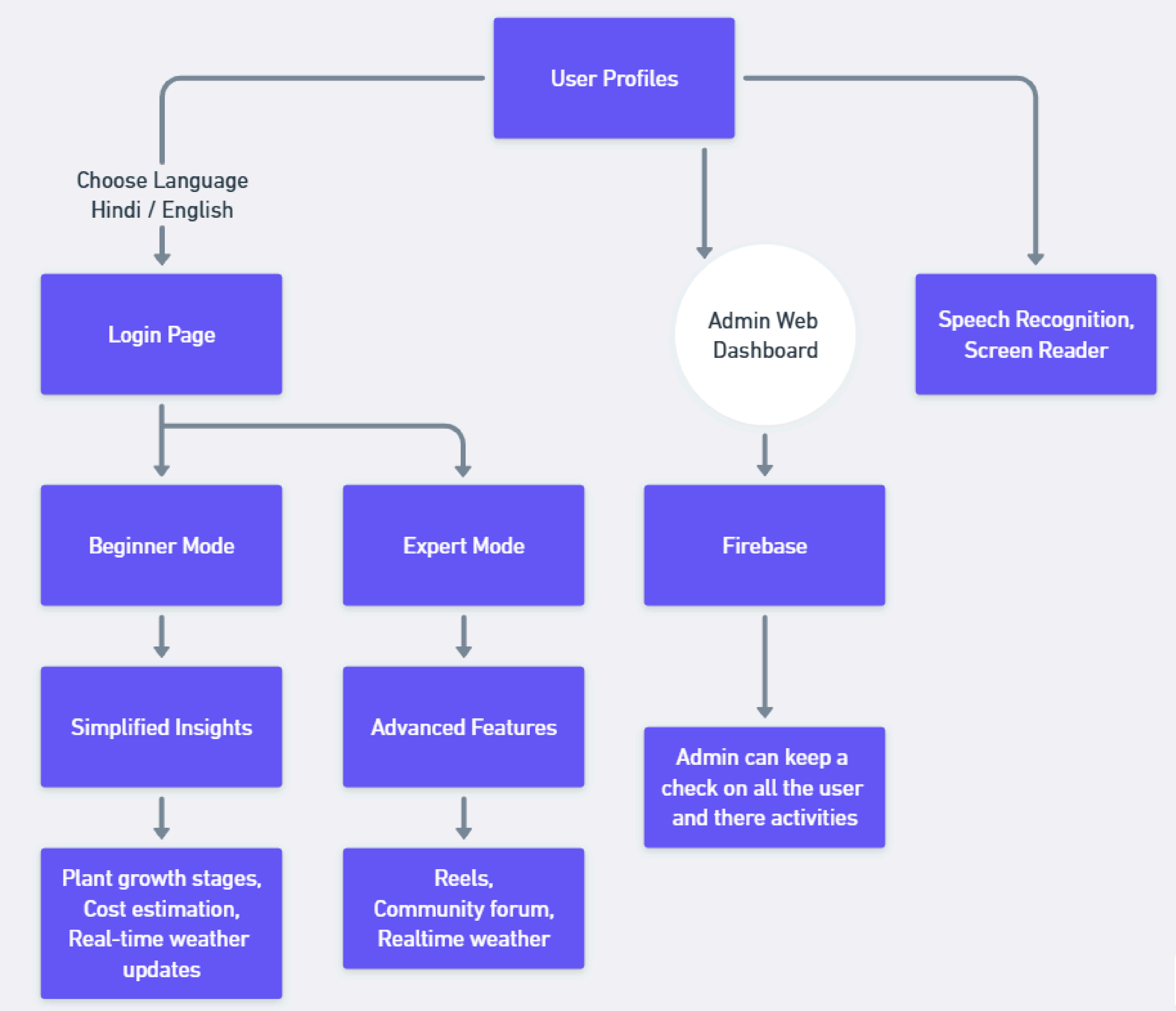
2. Problem Statement

Water scarcity is a pressing global challenge. Despite numerous water-saving techniques, the absence of a centralized repository for knowledge sharing hinders the widespread adoption of sustainable practices. Farmers and communities lack access to localized, actionable insights that could optimize water usage and prevent resource depletion. Existing platforms are fragmented, leaving valuable knowledge siloed. A holistic, interactive, and multilingual solution is needed to bridge this gap.

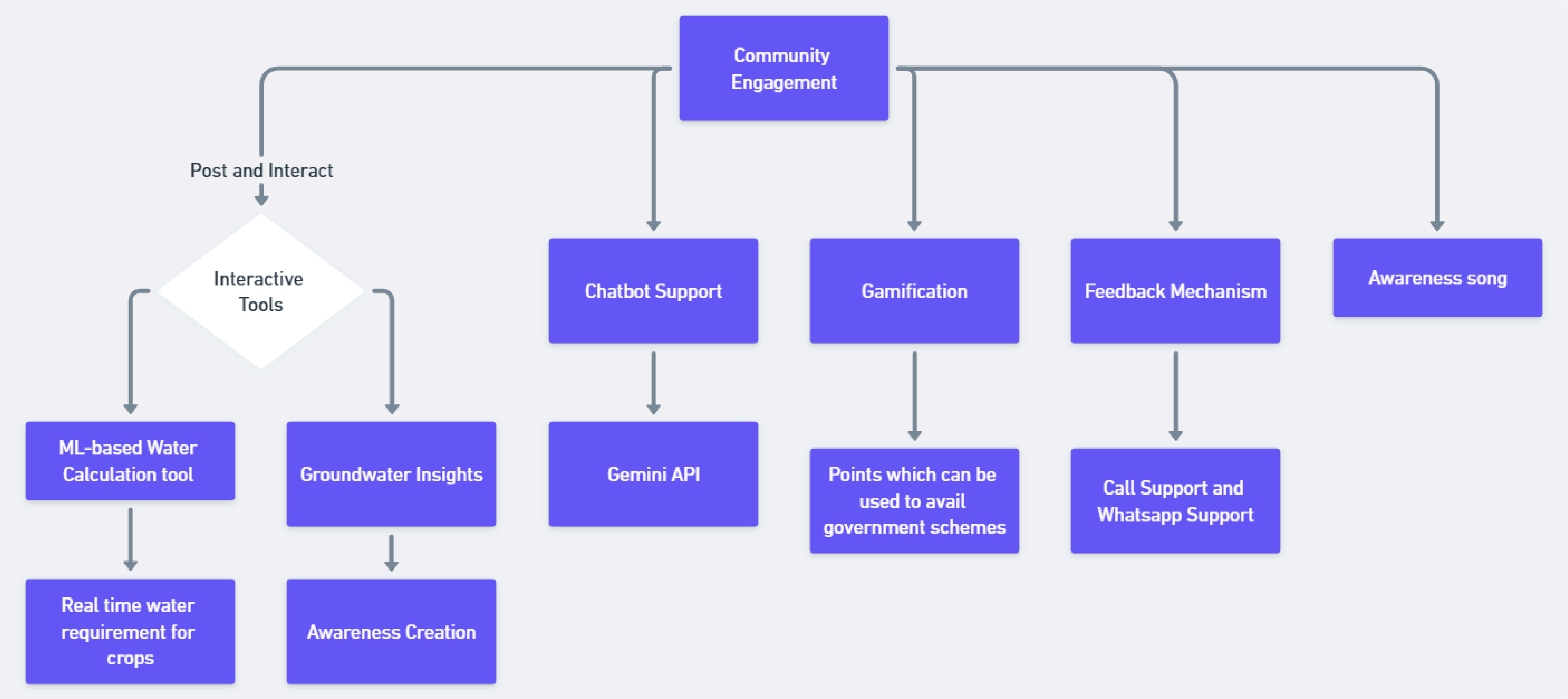
3. Solution Description

Our solution is a mobile-first, multilingual, wiki-style knowledge-sharing platform combined with machine learning models, gamification, and community features to drive water conservation awareness and action. The app is accessible to beginners and experts, providing customized insights, educational content, and interactive tools. The platform's key features include:

- **User Profiles**: Users can register, log in, and choose a language (e.g., **Hindi or English**). User data is stored securely on Firebase and monitored via a React-based admin web dashboard.
- **Beginner Mode**: Simplified insights like plant growth stages, cost estimation, and real-time weather updates.
- **Expert Mode**: Advanced features like water-saving techniques, research articles, community forums, and video content.
- **Community Engagement:** Users can post and interact with reels, articles, and discussions. Features include likes, comments, and reporting inappropriate content for moderation.



- **Interactive Tools**: ML-based irrigation calculators for Karnataka that predict daily water needs of crops using soil type, irrigation methods, and real-time weather data.
- **Groundwater Insights**: Groundwater availability data is visualized on an interactive map to educate users about resource management.
- **Assistive Technology**: Screen readers and speech recognition ensure accessibility for differently-abled users.
- **Chatbot Support**: A Gemini API-powered chatbot to address user queries.
- **Gamification:** Points earned in mini-games can be redeemed for government schemes, encouraging user participation.
- **Awareness Creation**: A custom water-saving song and visual storytelling to enhance engagement.
- **Feedback Mechanism**: A support system for user feedback and troubleshooting.



4. Summary

The platform integrates data, technology, and community-driven content to revolutionize water conservation efforts. By leveraging real-time weather data, ML models, and engaging user experiences, the solution provides a unique, actionable approach to tackle water scarcity.

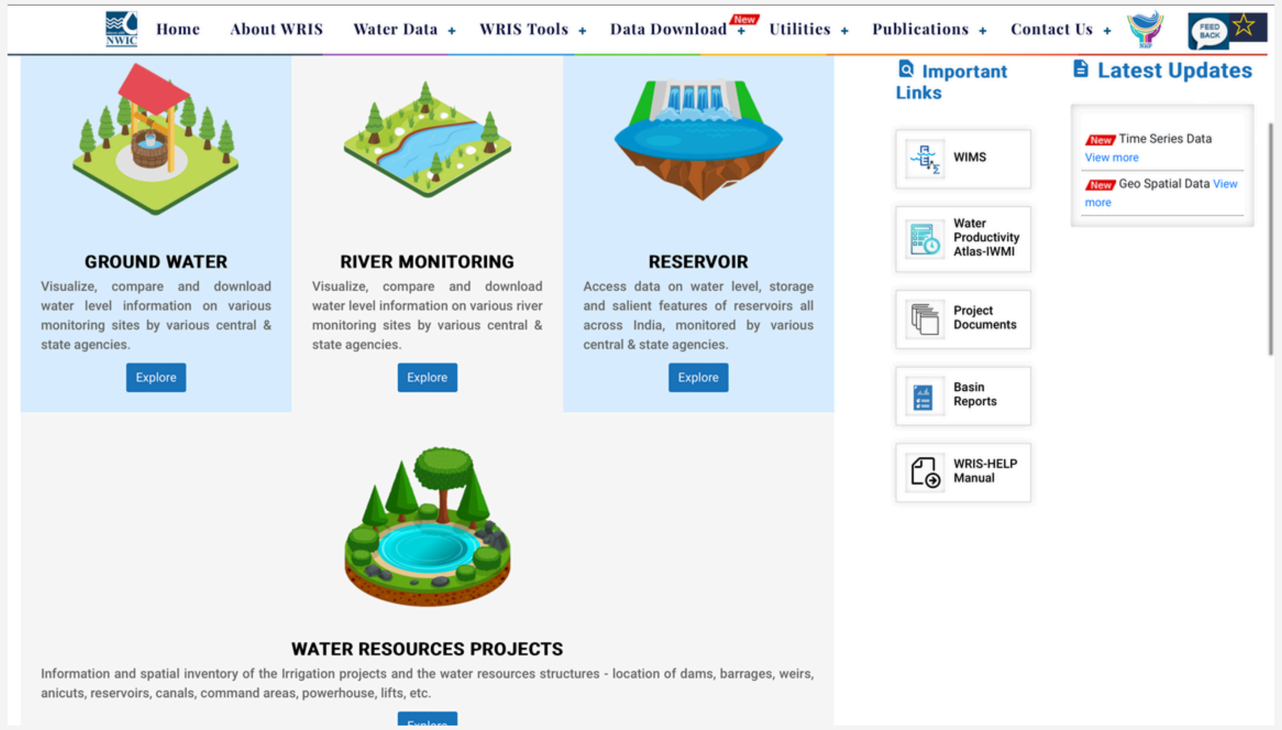
5. Technology Stack

- **Frontend**: **Flutter (mobile app)**, **React.js (admin dashboard)**
- **Backend**: Firebase for authentication and database management
- **ML Models**: Python (XGBoost for evapotranspiration prediction, soil moisture models) using flask deployed the ML model on Heroku
- **APIs**: Real-time weather data, Gemini API (chatbot)
- **Map Integration**: Open Street Map India
- **Assistive Tech**: Speech recognition, screen reader libraries

6. Solution Approach

- Data Collection: Extracted evapotranspiration and groundwater data from Ministry of Jal Shakti and India WRIS databases.
- Data Preprocessing: Cleaned and prepared datasets for machine learning models.
- ML Model Development:
  - Predicted daily irrigation needs for specific crops using factors like soil type, weather data, and irrigation methods.
  - Incorporated groundwater availability to provide location-specific insights.
- App Development:
  - Designed user-friendly interfaces for multilingual support.
  - Integrated gamification, chatbot, and assistive features.
- Testing: Ensured usability and accuracy of ML predictions through iterative testing and checked every functionality of app.
- Deployment: Hosted the app and admin dashboard for seamless user and admin experiences.

WRIS

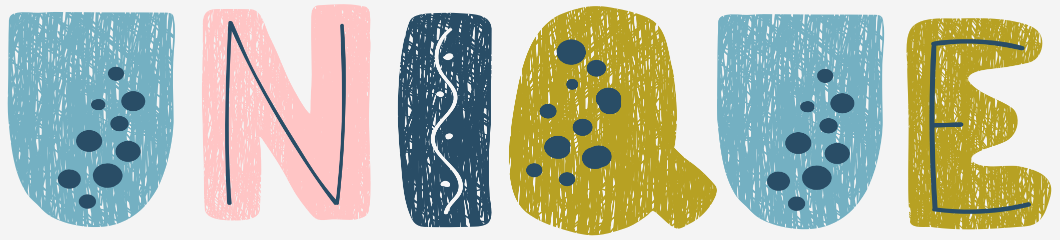


7. Theme- Clean and Green Technology

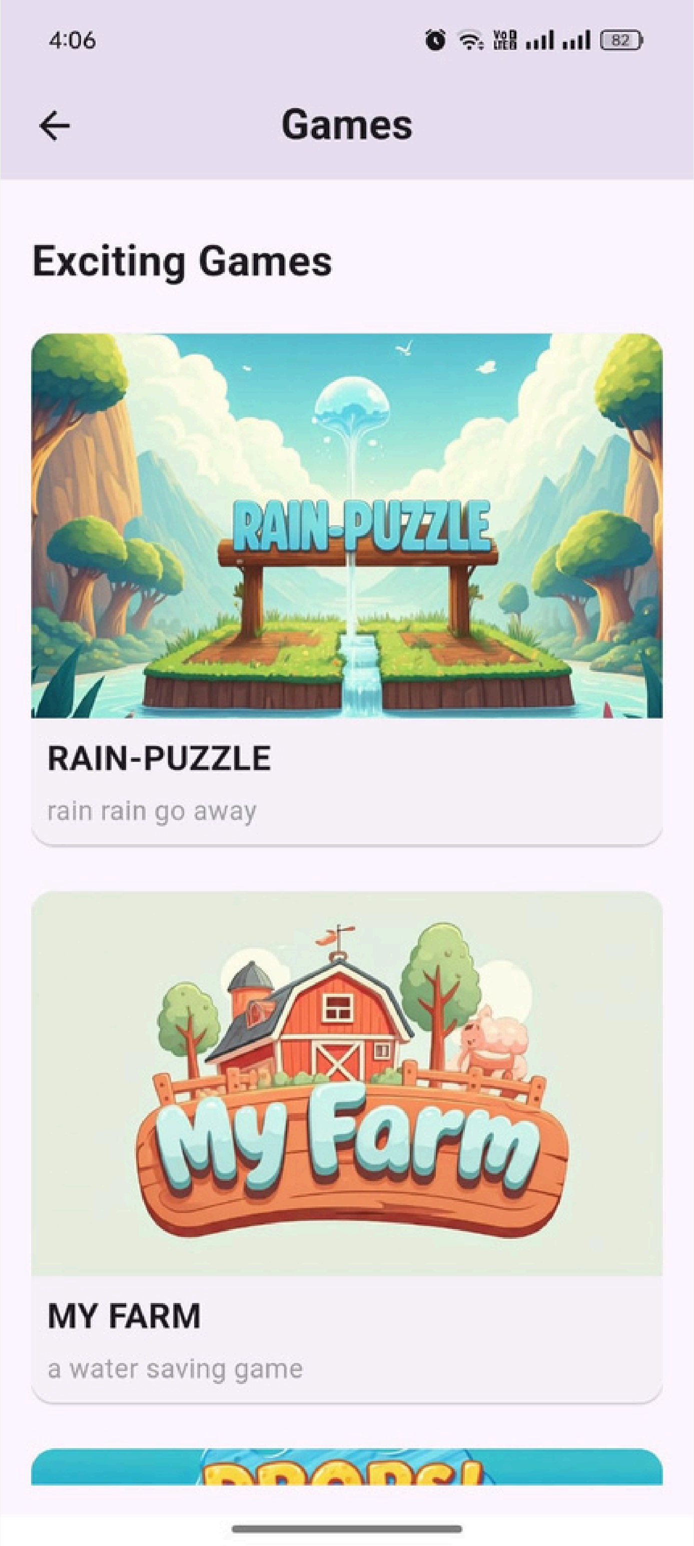
The theme of our project is "Water Conservation through Technology and Awareness." By combining community engagement, innovative tools, and actionable insights, the solution fosters sustainable practices and addresses the global challenge of water scarcity.

HOW DIFFERENT IS YOUR SOLUTION FROM EXISTING SOLUTION.

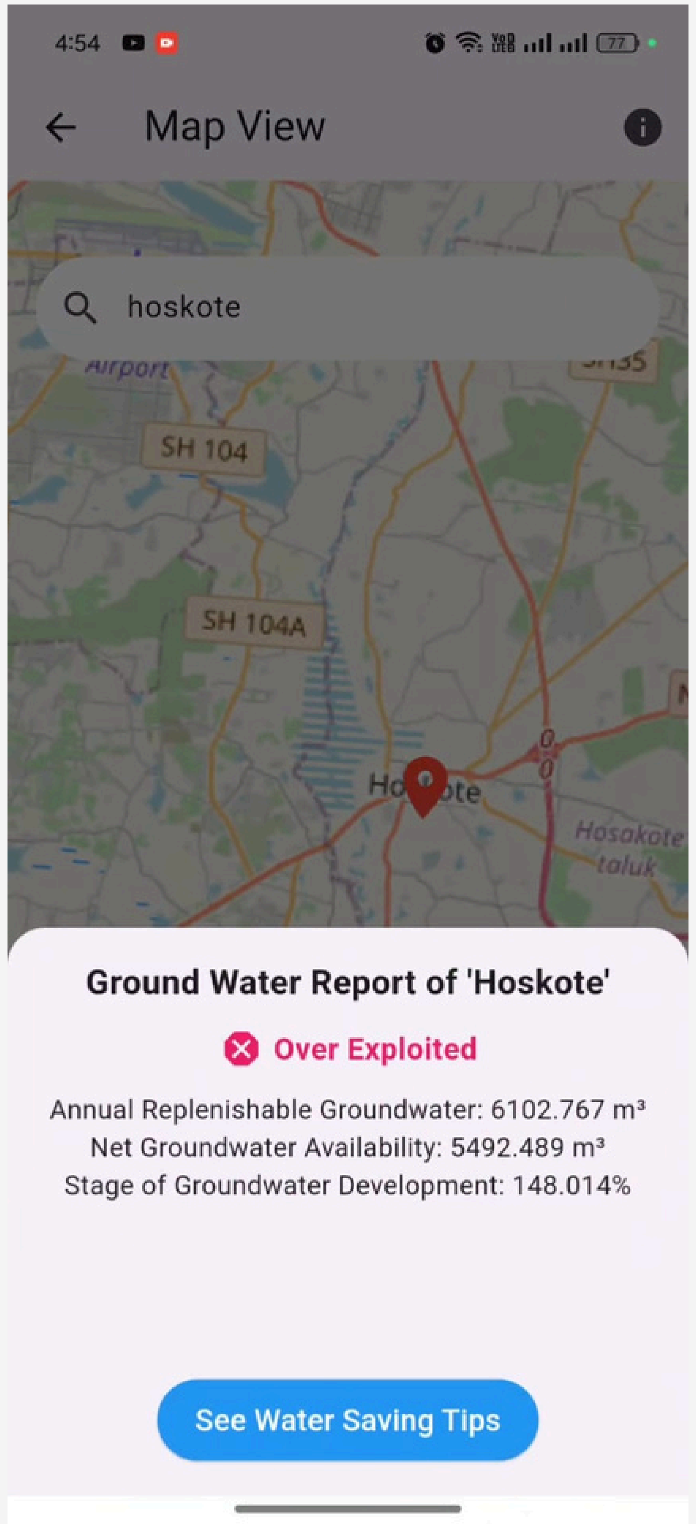
- **Centralized Repository:** Combines irrigation techniques, water-saving practices, farming news, and government schemes in one app, unlike fragmented resources on other platforms.
- **Localized Insights:** Provides machine learning-driven irrigation predictions tailored to specific locations, while others offer generic advice.
- **Groundwater Mapping:** Includes interactive maps showing local groundwater availability, which most platforms lack.
- **Multilingual & Accessible:** Offers Hindi and English support with speech recognition and screen readers, ensuring inclusivity.
- **Community Engagement:** Features reels, articles, and forums for user interaction and knowledge sharing, unlike one-way information from other platforms.
- **Search & Voice Assistant:** Offers voice search and keyword-based search in any language, making it more accessible than traditional search functions.
- **Real-Time Weather Updates:** Provides live weather data for irrigation, unlike platforms without real-time integration.
- **Gamification:** Rewards users with points that can be redeemed for government schemes, encouraging active participation.
- **Support & Feedback:** Includes contact support via call/WhatsApp and a feedback system, unlike many platforms with limited support.



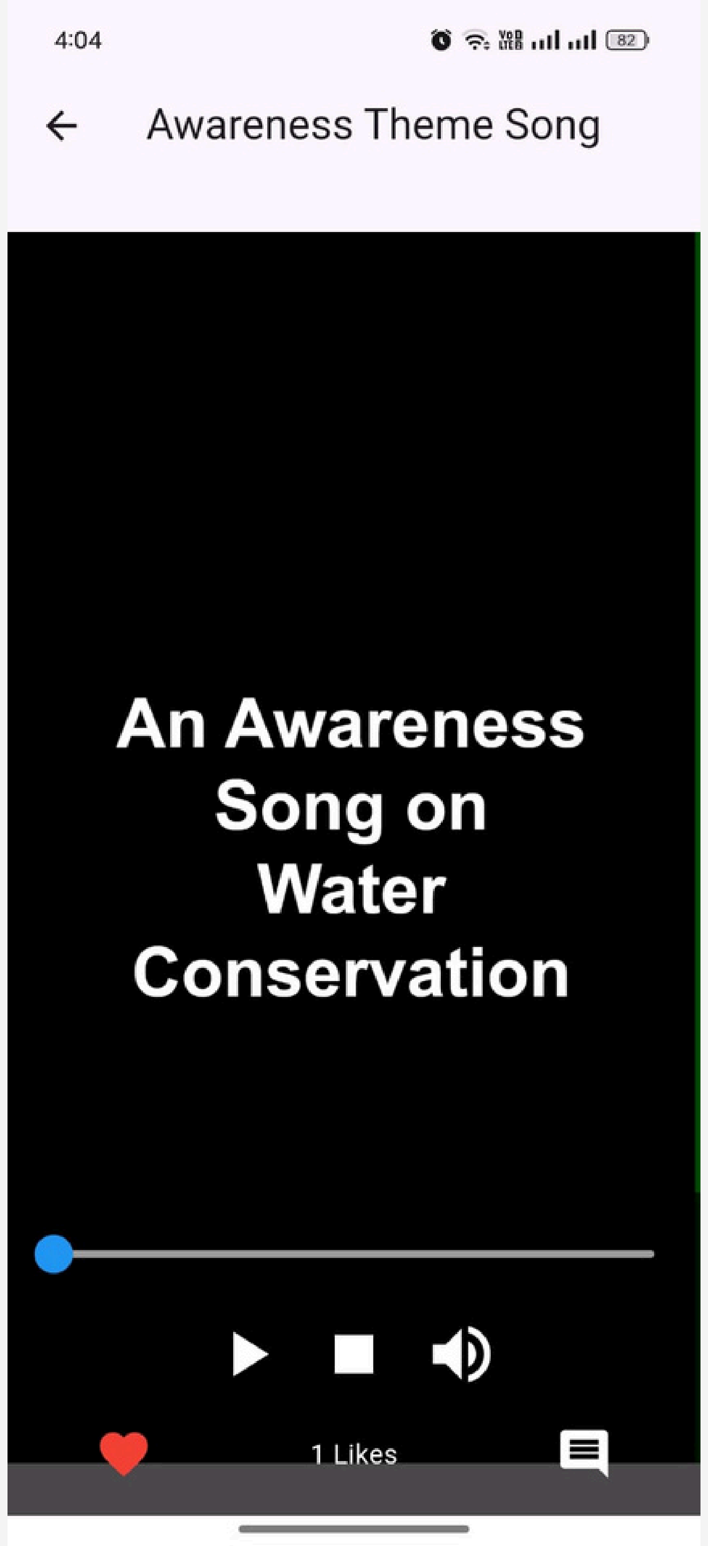
NOVEL FEATURES



Gamification for Awareness: Points and rewards gamify water conservation education.

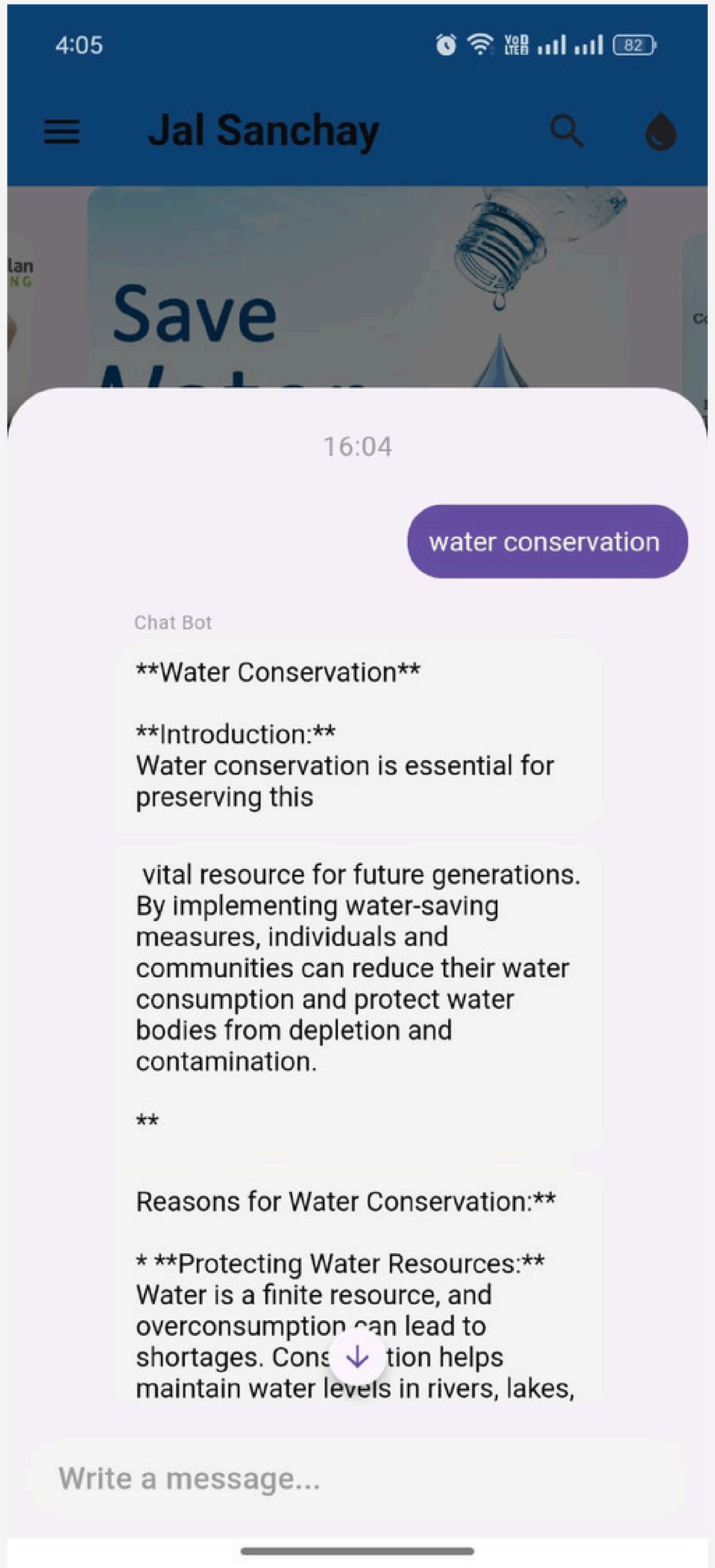


Groundwater Mapping: Visual groundwater availability insights empower users to make informed decisions.

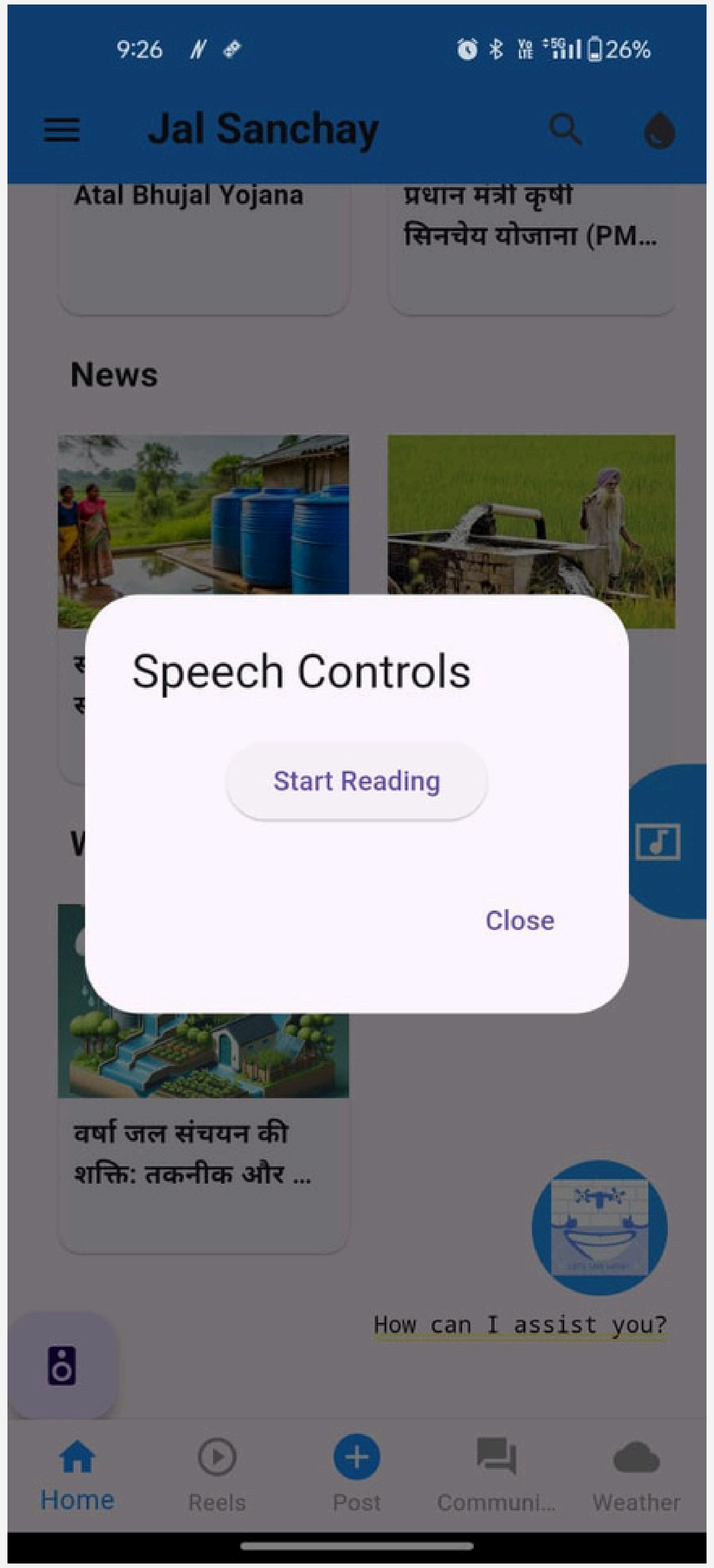


Awareness Through Creativity: A custom water-saving song and visual tools reduce reliance on text-heavy content.

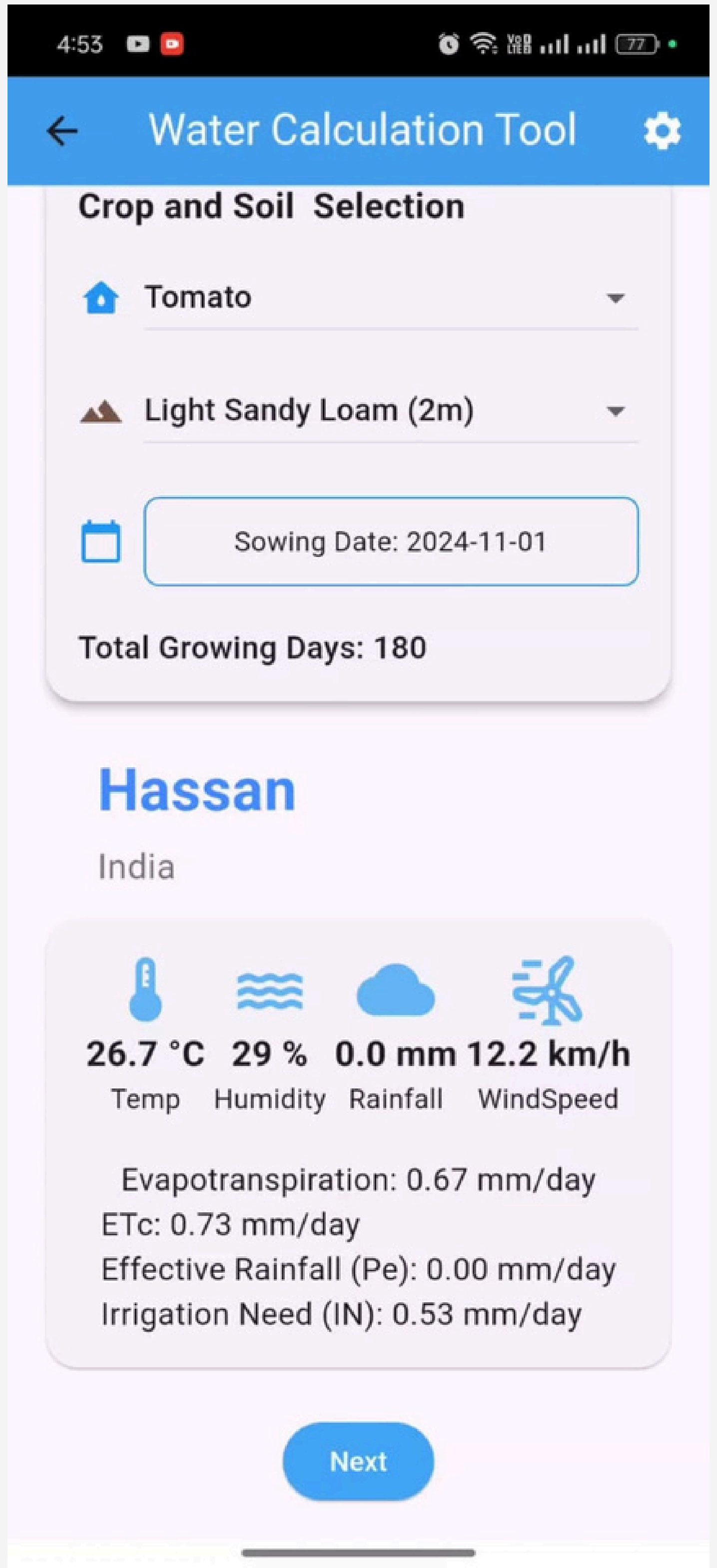
NOVEL FEATURES



Chatbot Support: A Gemini API-powered chatbot to address user queries.



Accessible and Inclusive: Accessible and Inclusive: Screen readers and speech recognition ensure usability for all, including visually impaired users.

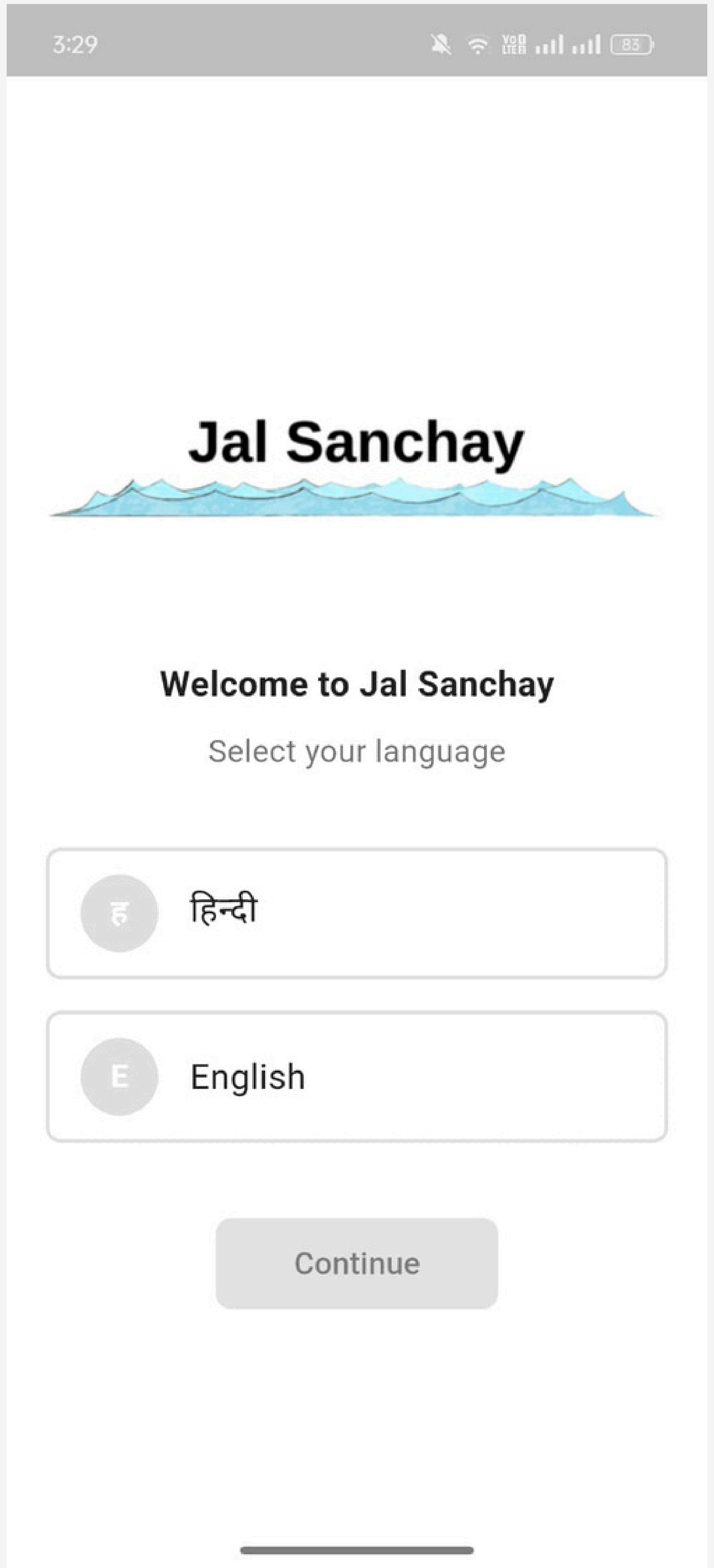


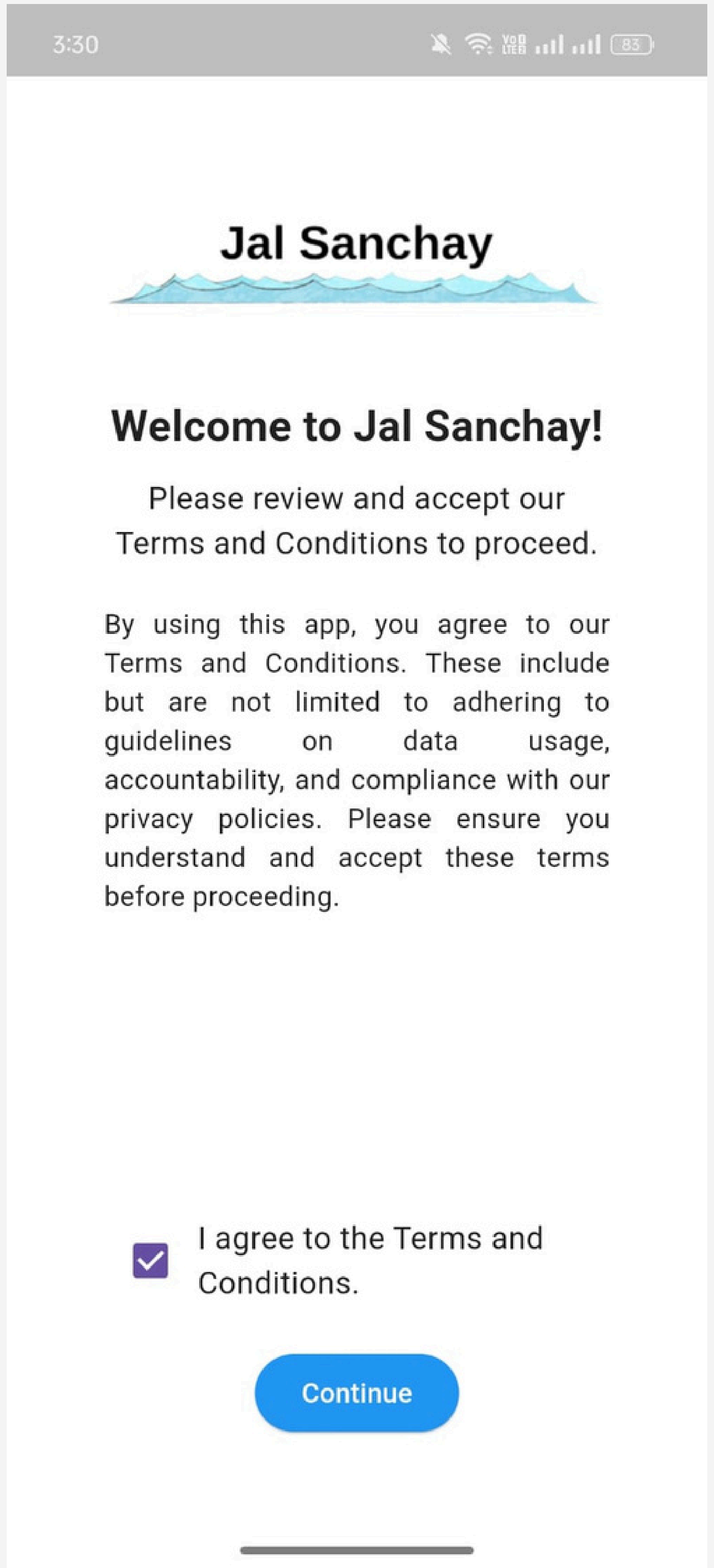
Localized Insights: ML models for Karnataka provide precise, real-time irrigation recommendations based on location-specific data.

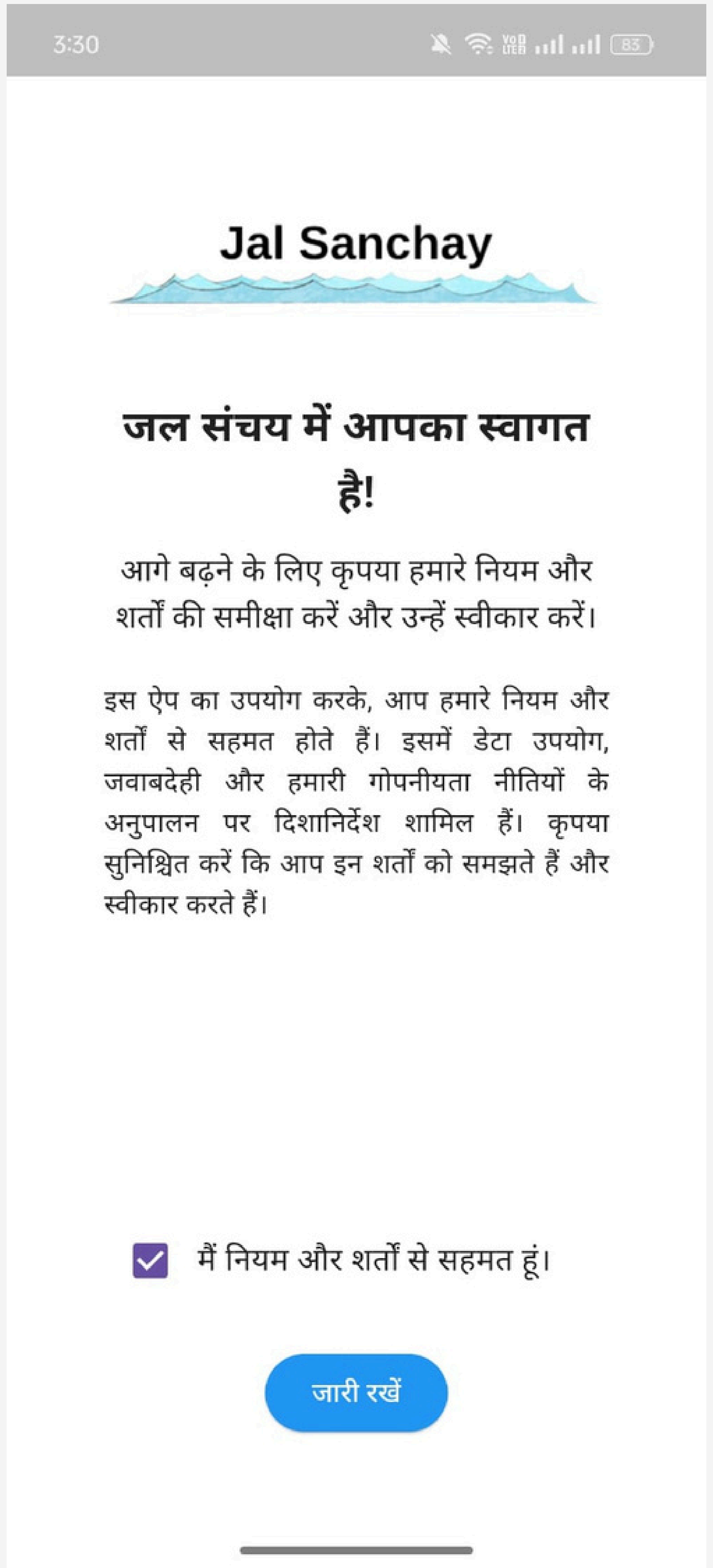
Jal Sanchay

SNAPSHOTS OF APP

LANDING PAGE

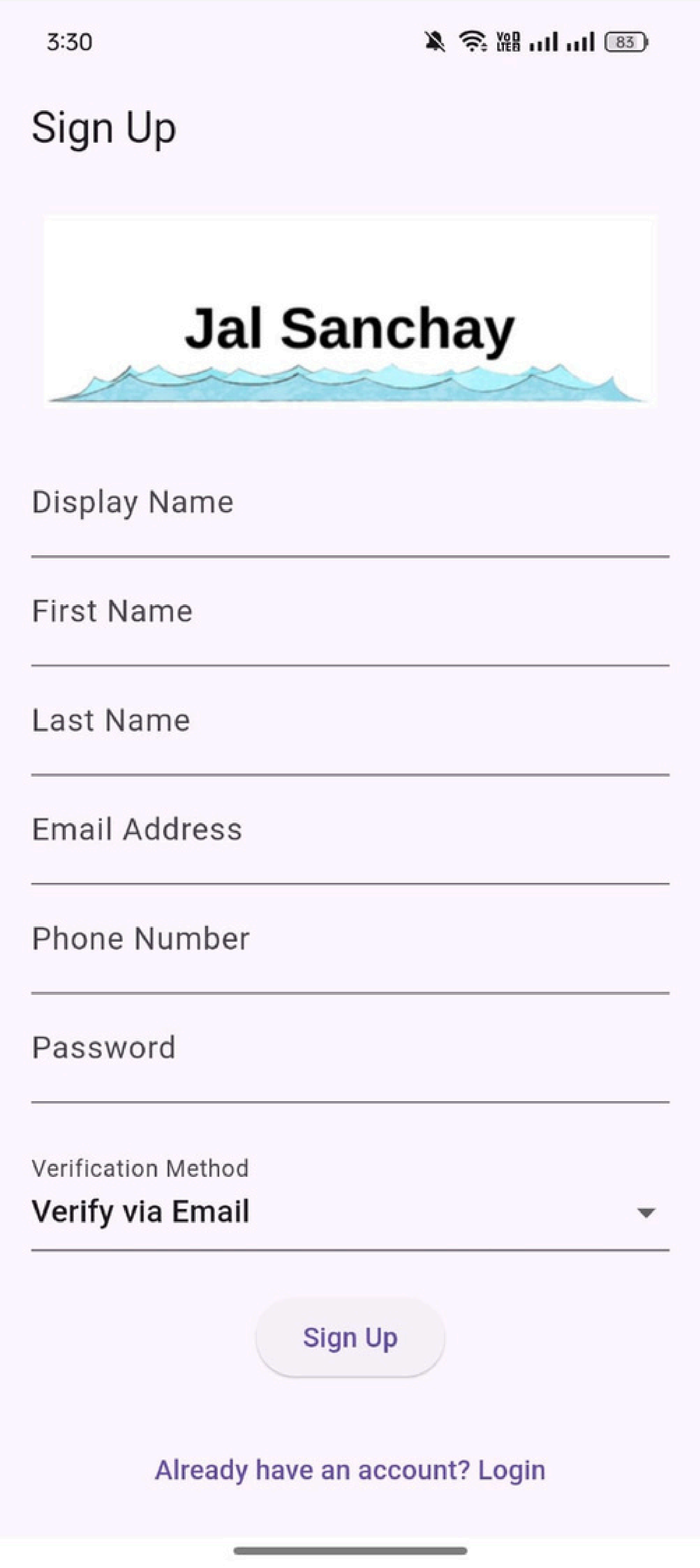


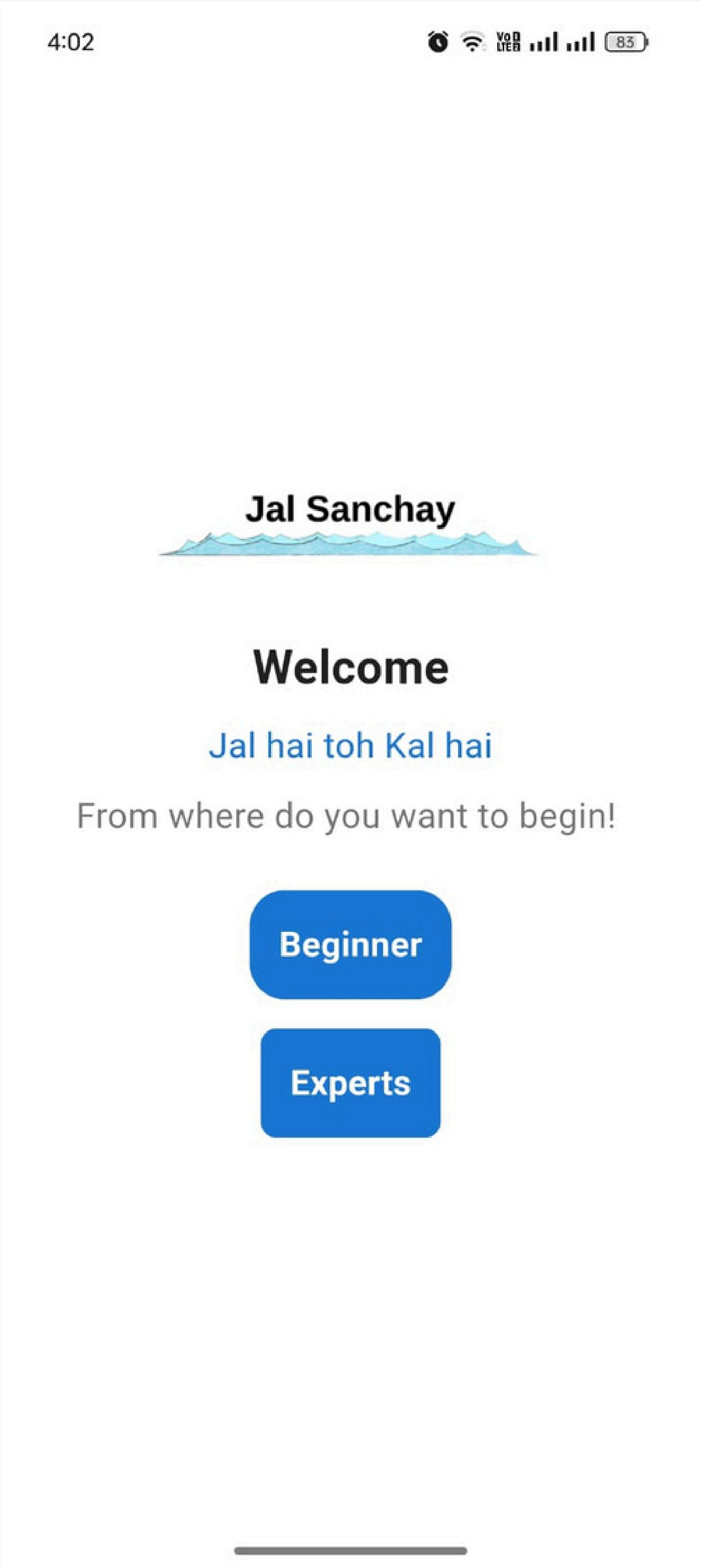




- **Multilingual Home Page:** The app starts with an option to select the preferred language (Hindi or English).
- **Language-Specific Terms and Conditions:** After selecting a language, the terms and conditions page is displayed in the chosen language for user agreement.

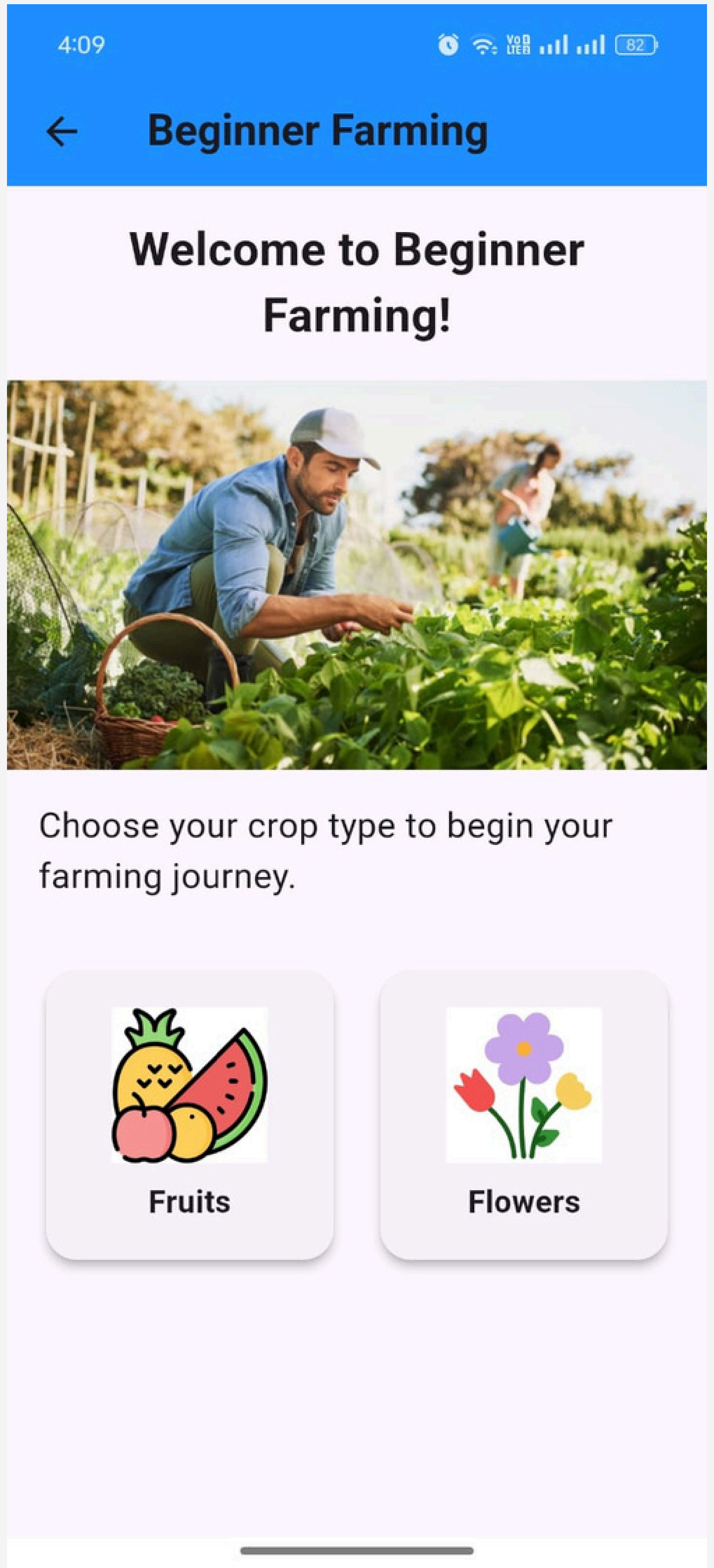
SIGN UP AND USER JOURNEY PAGE

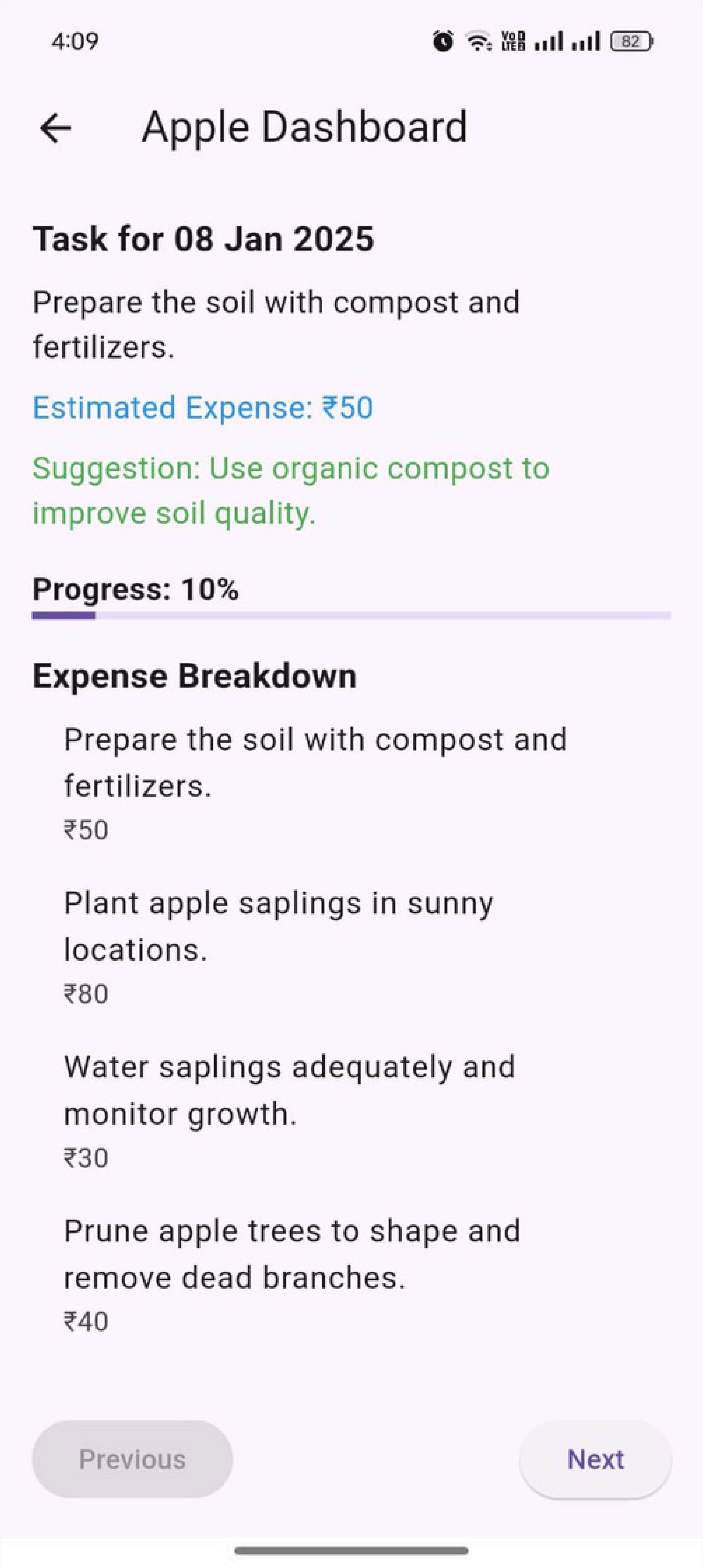


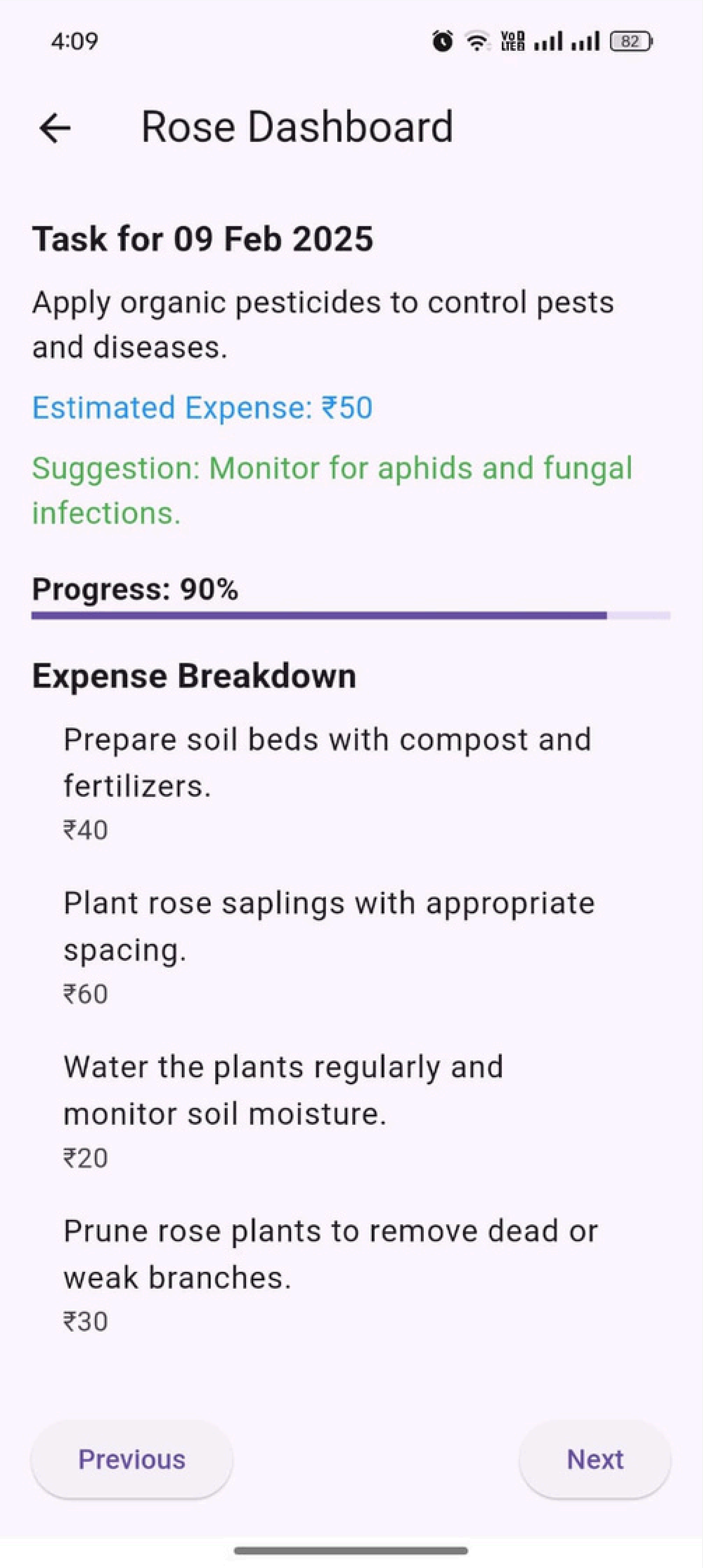


- **Signup Page:** Users can create a new account by entering details such as username, email, and password, ensuring secure access to the app.
- **User Journey Page:**
  - **For Beginners (Home Gardeners):** Tailored guidance on growing plants at home, including plant growth stages, care tips, overall expenditure and water-saving methods.
  - **For Beginners (Farmers):** Insights and tools to optimize crop irrigation, minimize water wastage, and implement efficient water conservation techniques.

USER JOURNEY PAGE FOR BEGINNERS & DASHBOARD



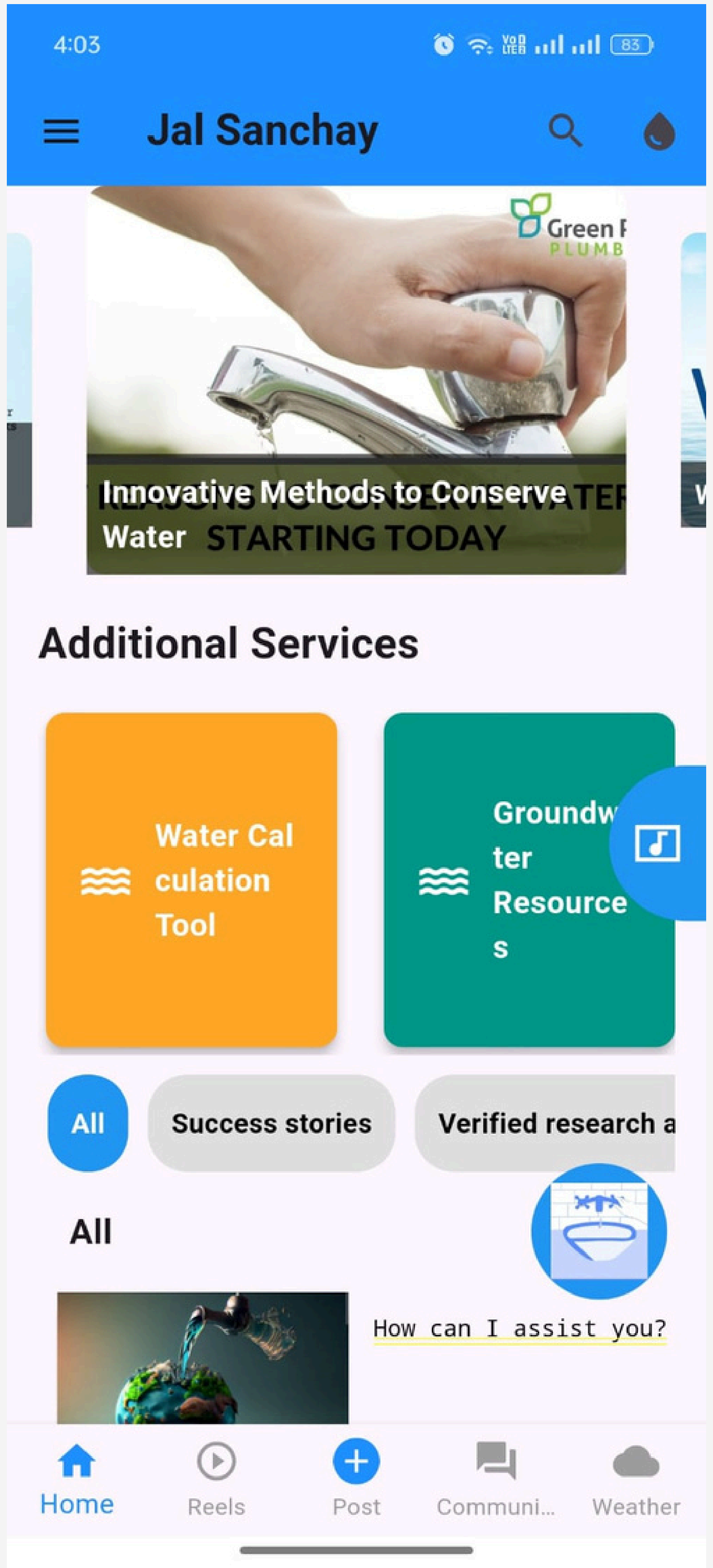


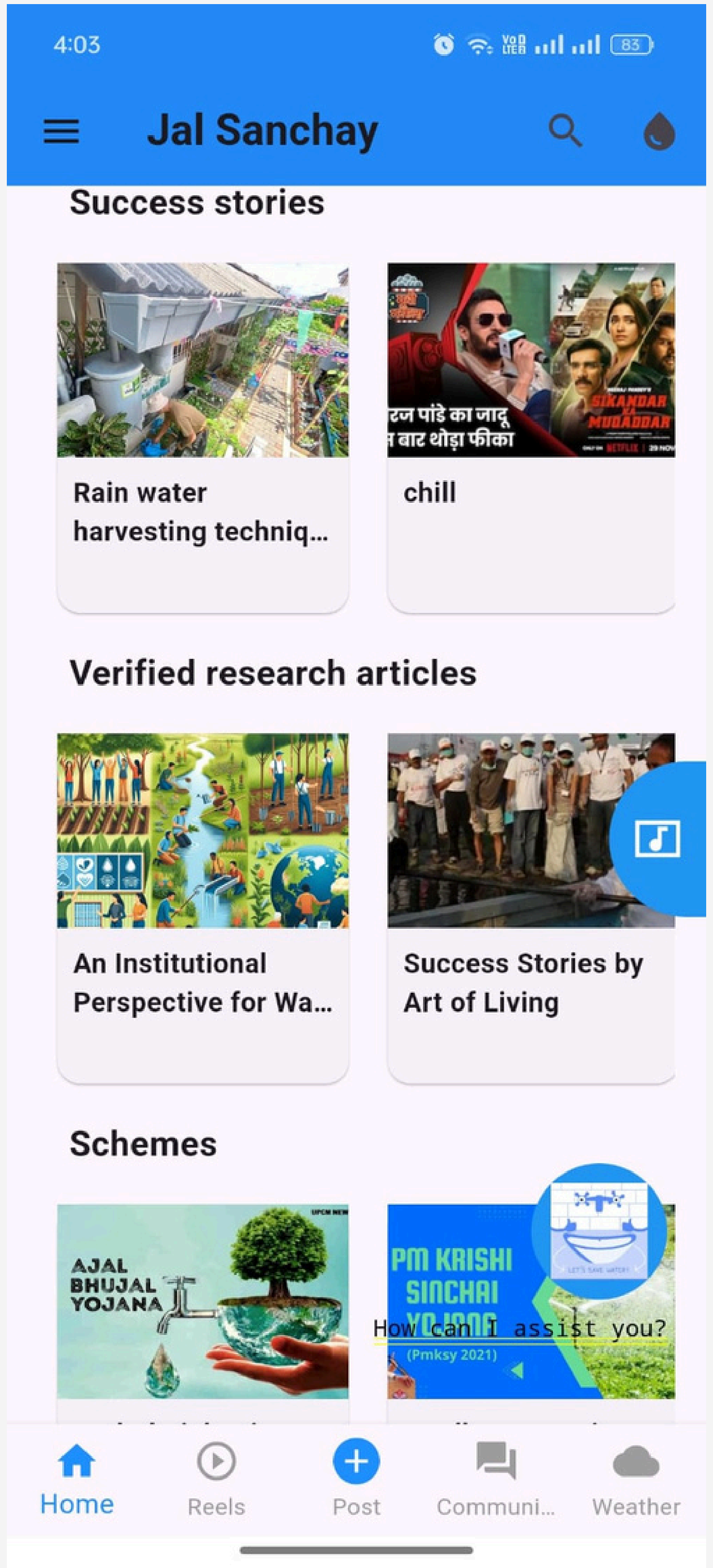


Home Page (Beginners) -
For those users who wants to learn about growing their favourites fruits or flowers at **home level** could use this page.
This page would help them in various ways as -

1. **learning** about their plant in an interactive way.
2. Getting **recommendations** on what they need to do for each day to help them grow their desired seed the best way possible.
3. To **avoid overspending** and understand what their estimated cost could be.
4. **Progress Tracking** is also available .

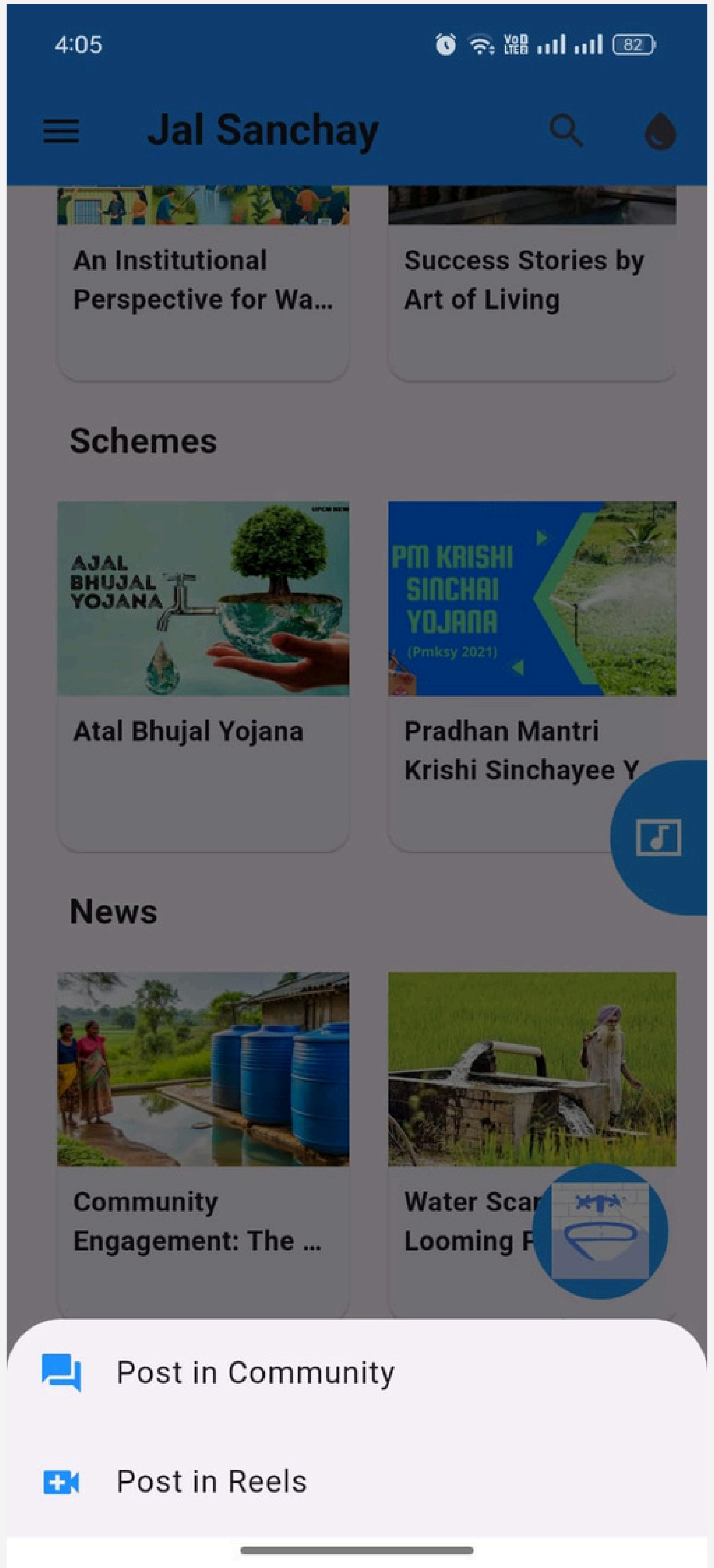
HOME PAGE- CENTRALIZED REPOSITORY(FOR EXPERTS)

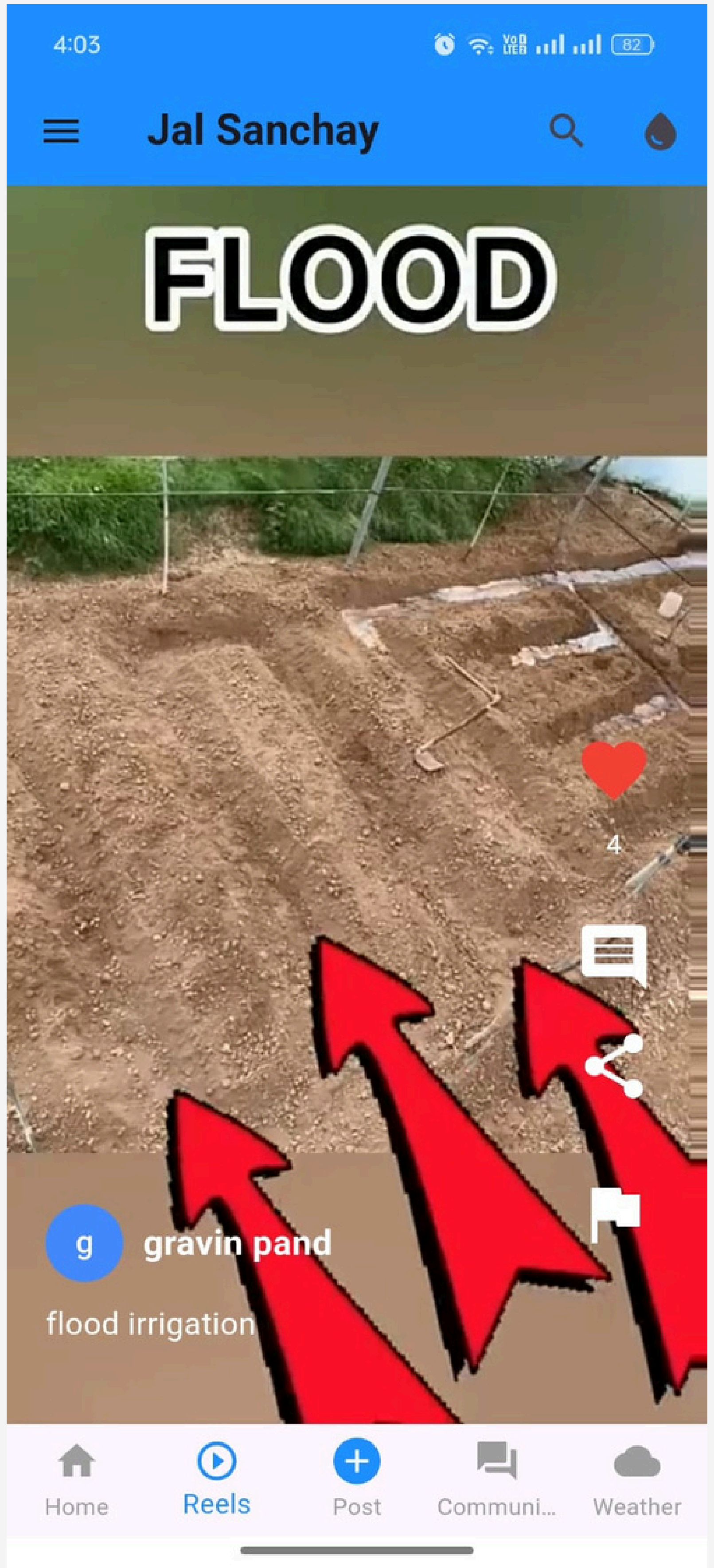


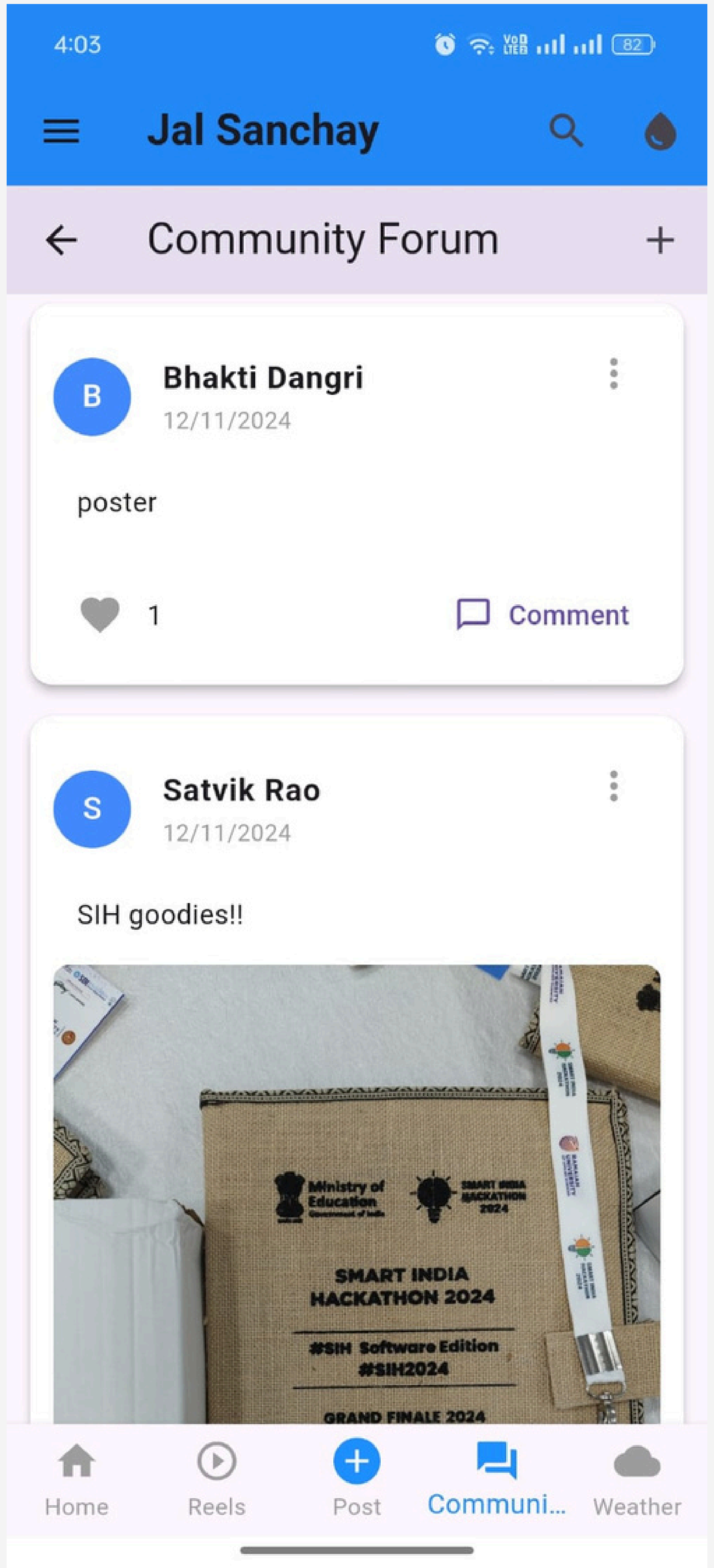


- **Home Page (Centralized Repository)**: A comprehensive hub providing verified information, including farming news, research articles on water-saving techniques, government schemes related to water conservation, and other agriculture-focused initiatives, ensuring users have access to reliable and up-to-date resources.

REELS AND COMMUNITY FORUM PAGE

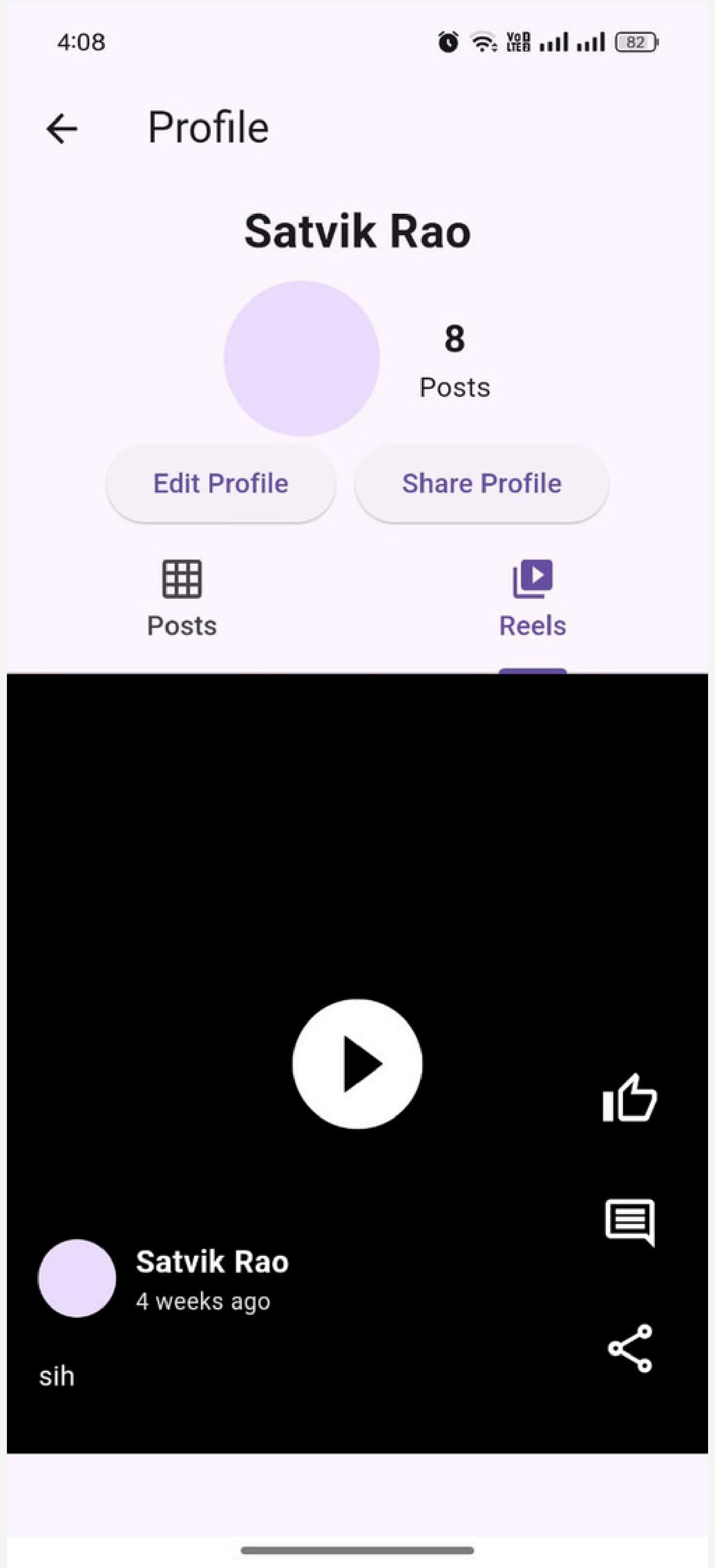


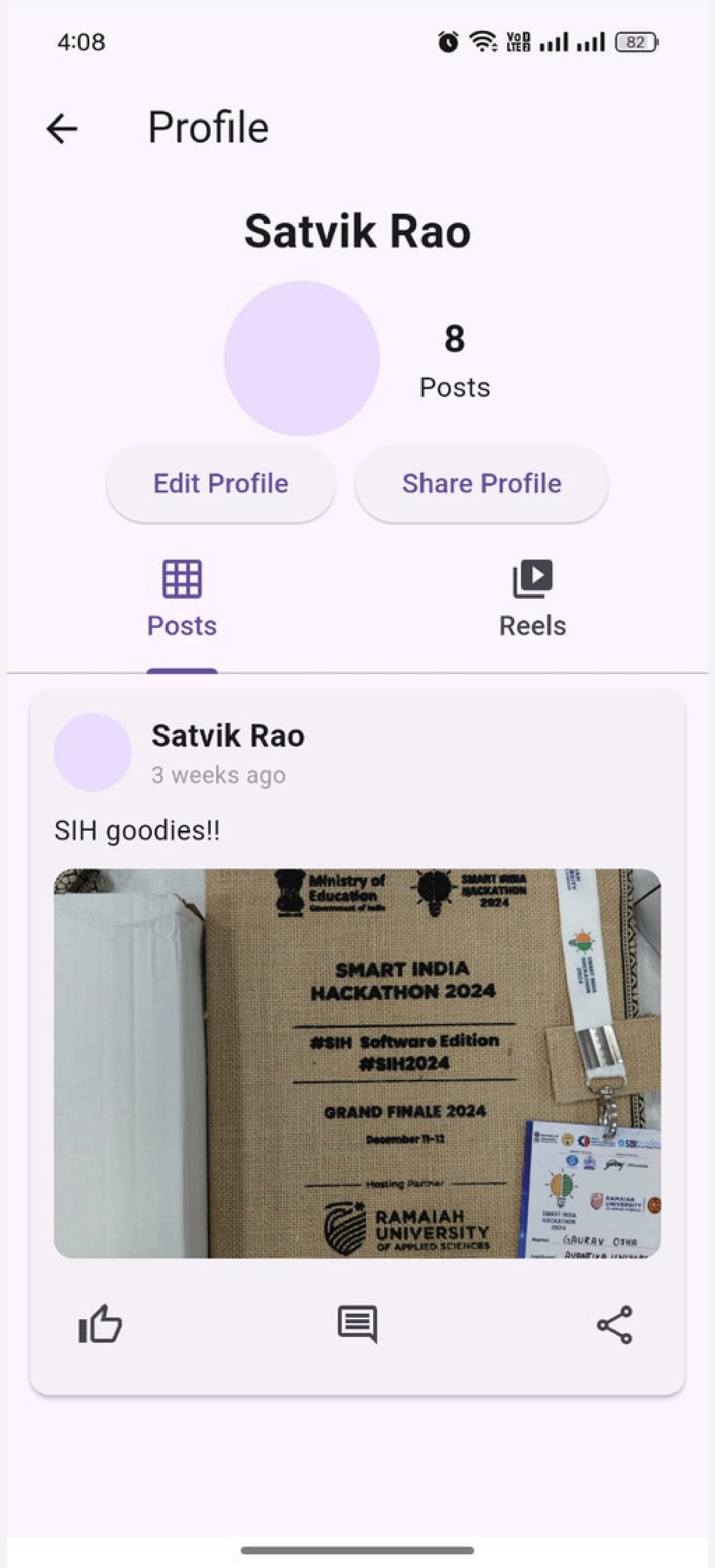




- **Reels Section:** Farmers can upload and share creative water-saving techniques and farming ideas through engaging reels, including practical "jugaadi" solutions.
- **Community Forum:** A space for farmers to connect, ask questions, share ideas, and collaborate to solve challenges and exchange knowledge.

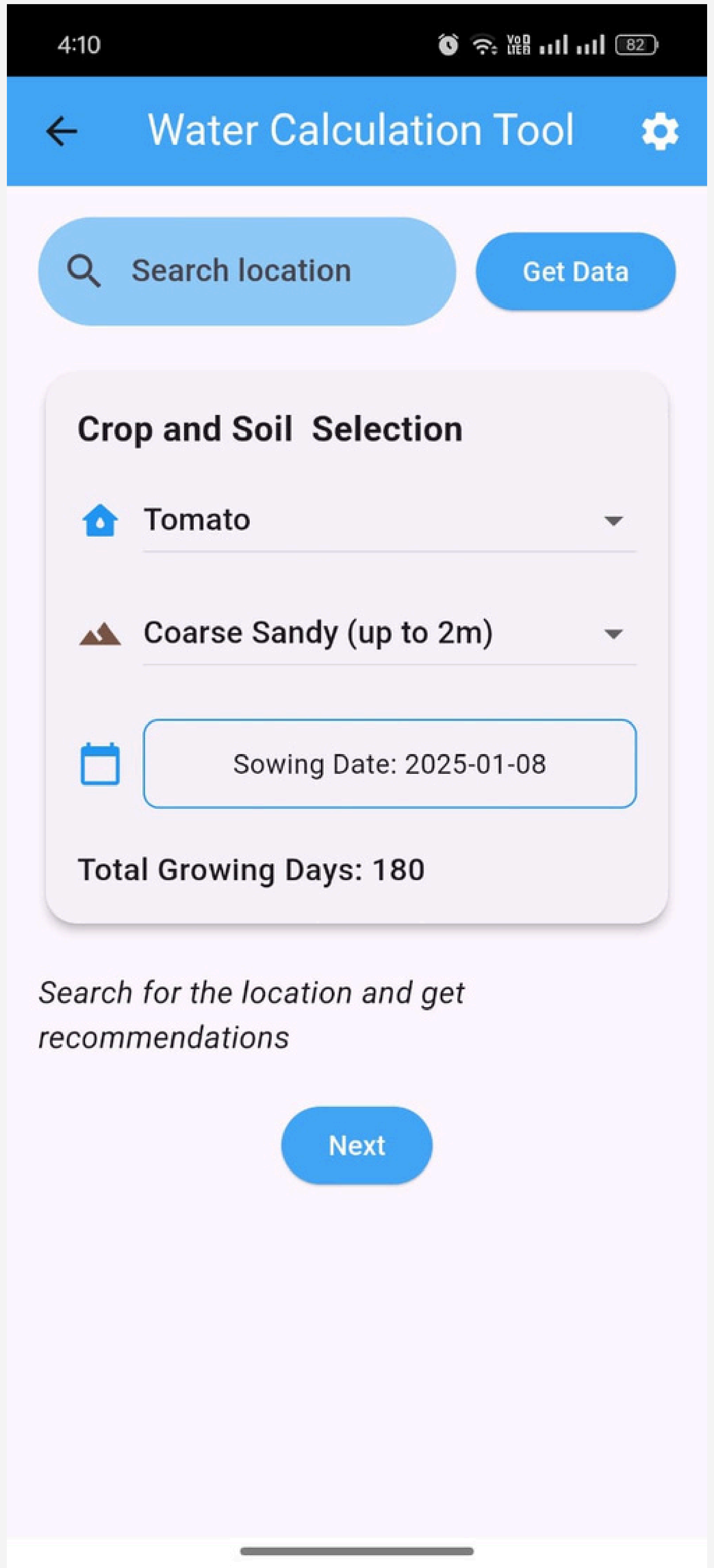
SEARCH AND PROFILE PAGE

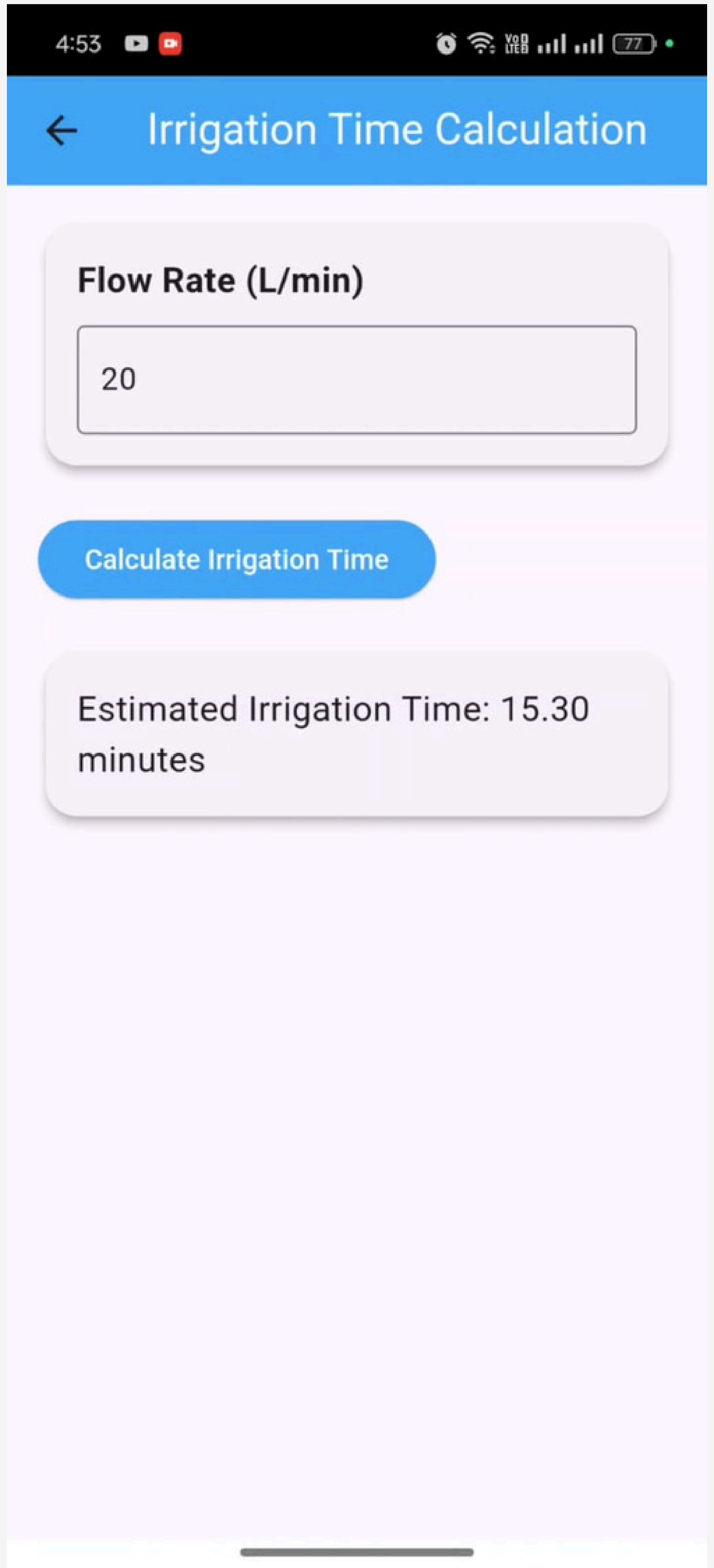




- **Search Page:** Farmers can search for content such as reels, community posts, articles, or schemes using keywords (e.g., "drip irrigation") or voice commands in their preferred language (Hindi/English), ensuring convenience and accessibility.
- **Profile Page:** Farmers can go through their posted reels and community posts in this page.

WATER CALCULATION TOOL & IRRAGITAION CALCULATOR

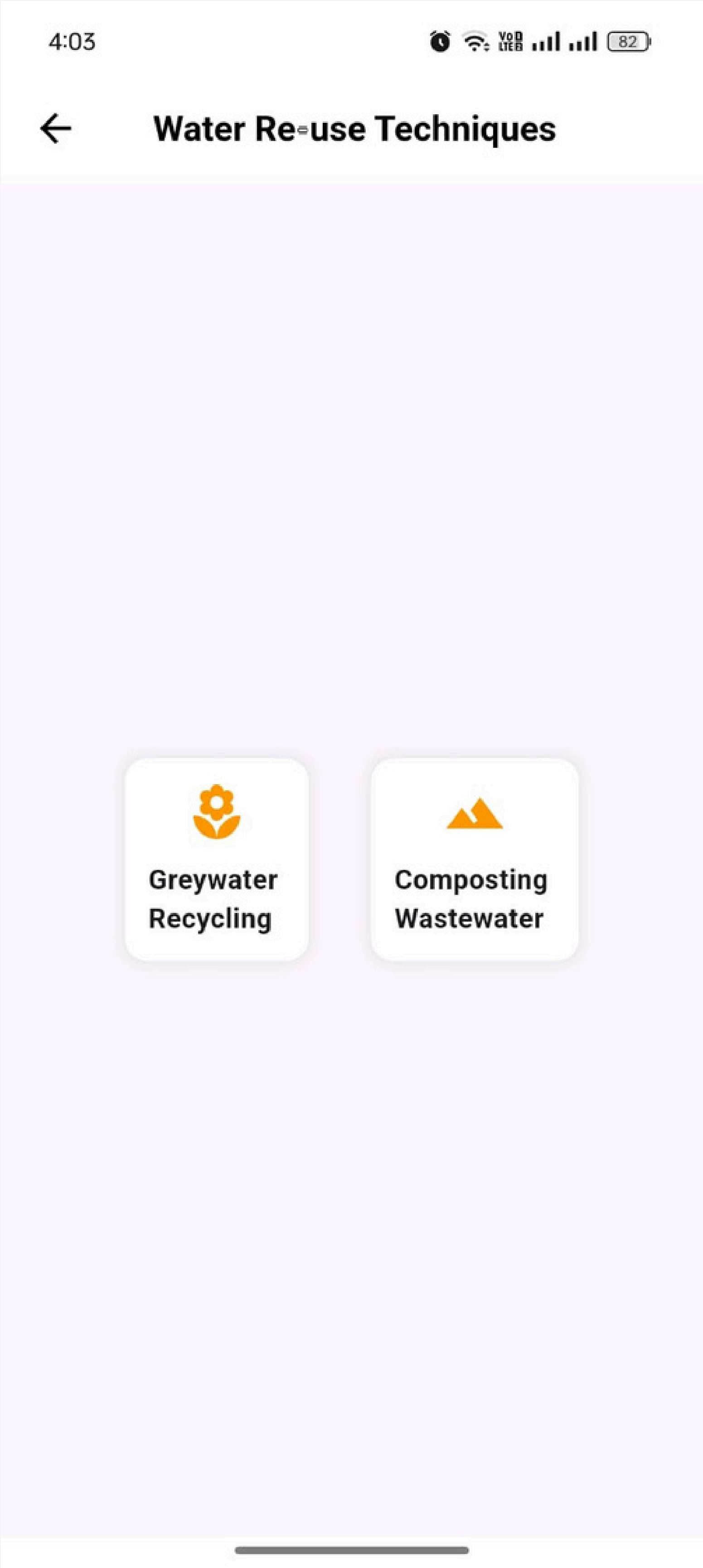


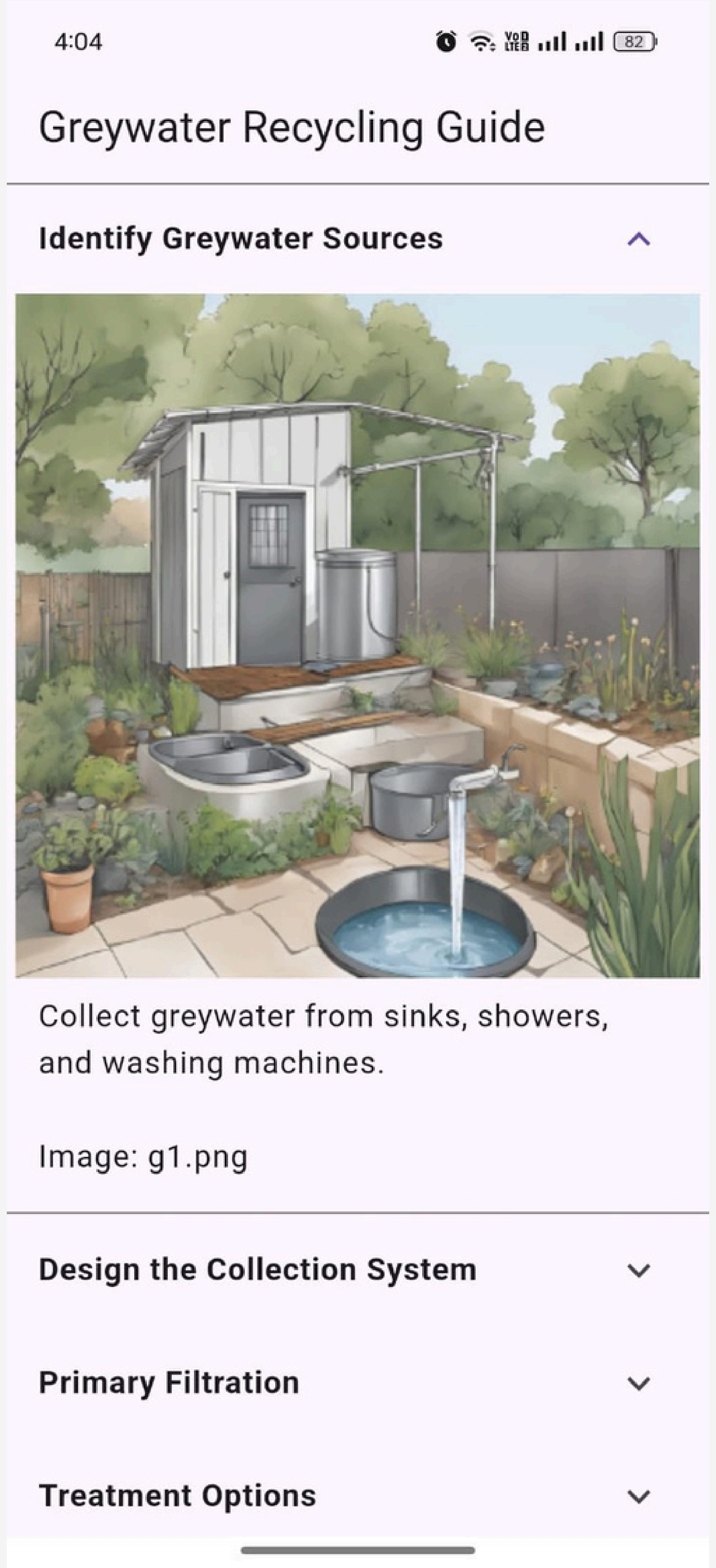


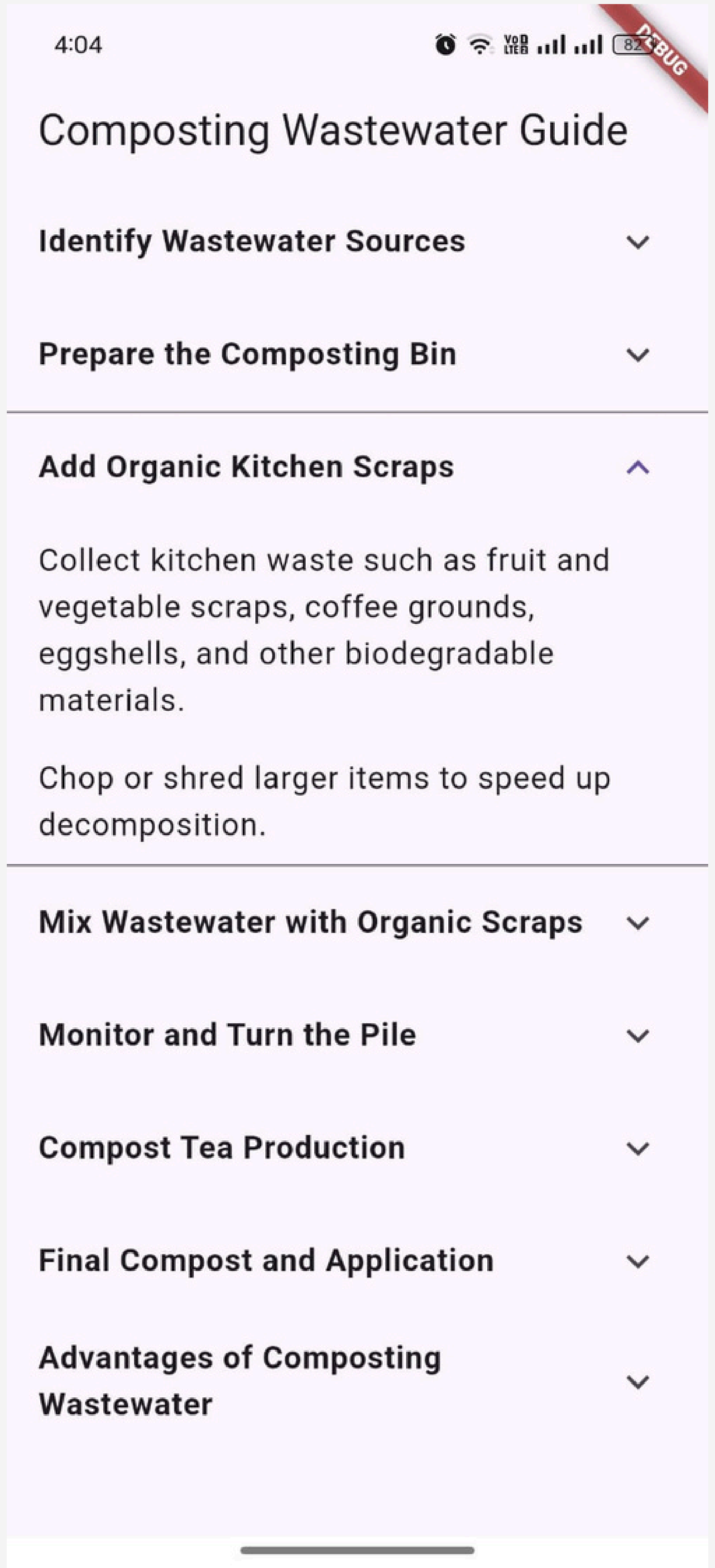
- **Water Calculation Tool Page:** Users can calculate the water requirements for their crops based on specific details such as crop type, soil type, and sowing date. Users input their crop and soil information, and the app provides tailored water requirement recommendations to ensure efficient water management. This page helps users efficiently manage their water resources by providing specific recommendations based on their crop and soil details.

- **Irrigation Time Calculation Page:** Users can calculate the time required to irrigate their fields based on the flow rate of their irrigation system. This helps in planning and optimizing irrigation schedules to ensure efficient water use. This page helps users determine the optimal irrigation time based on their system's flow rate, ensuring effective water management and reducing wastage.

WATER RE-USE TECHNIQUES

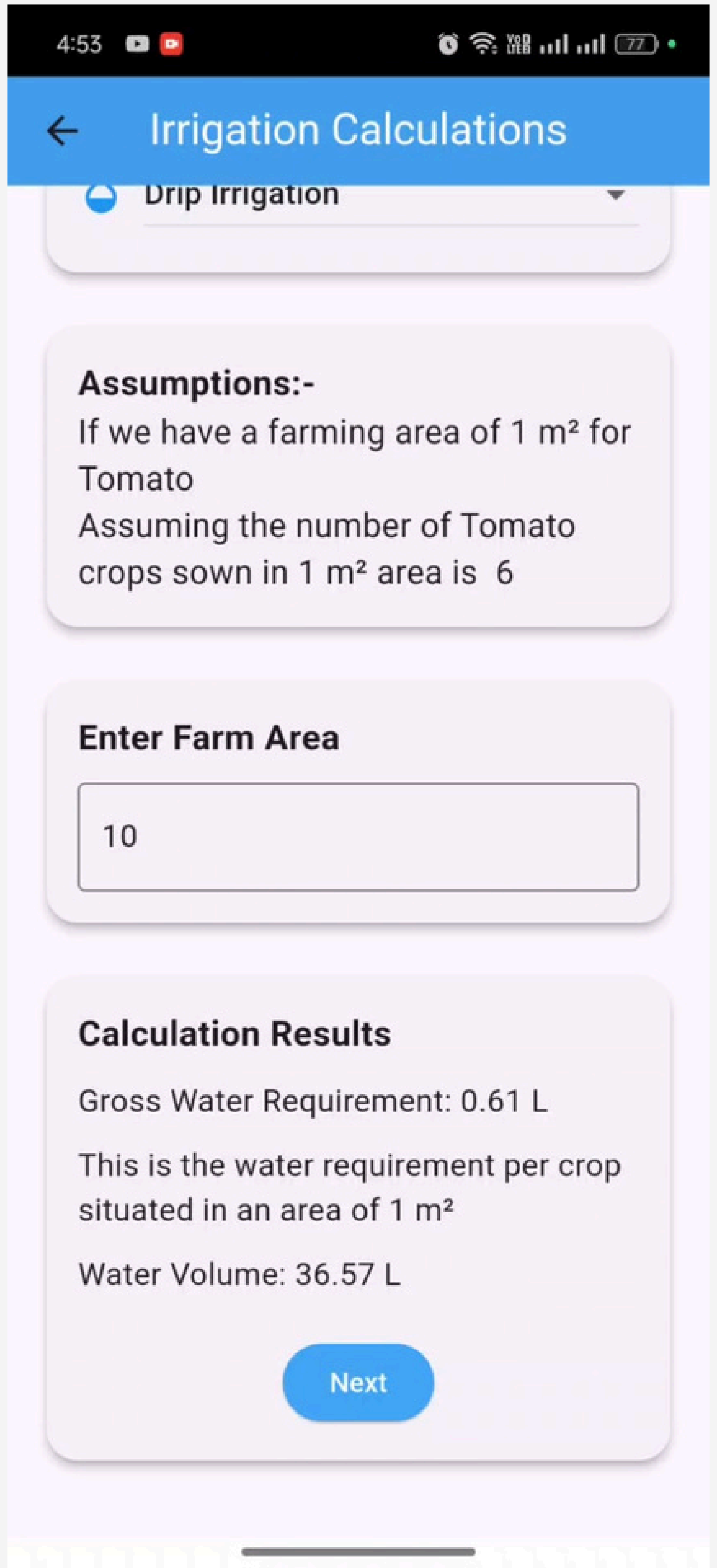


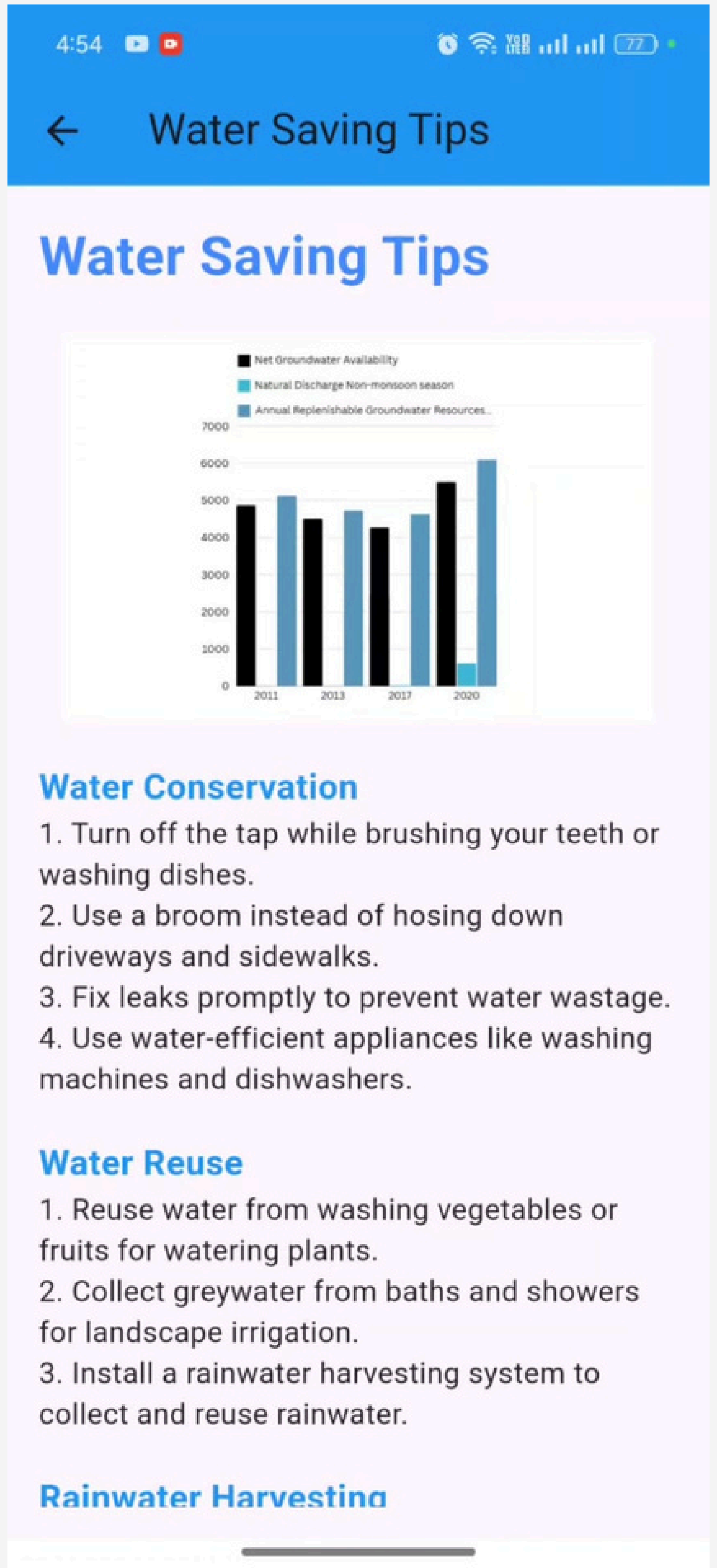


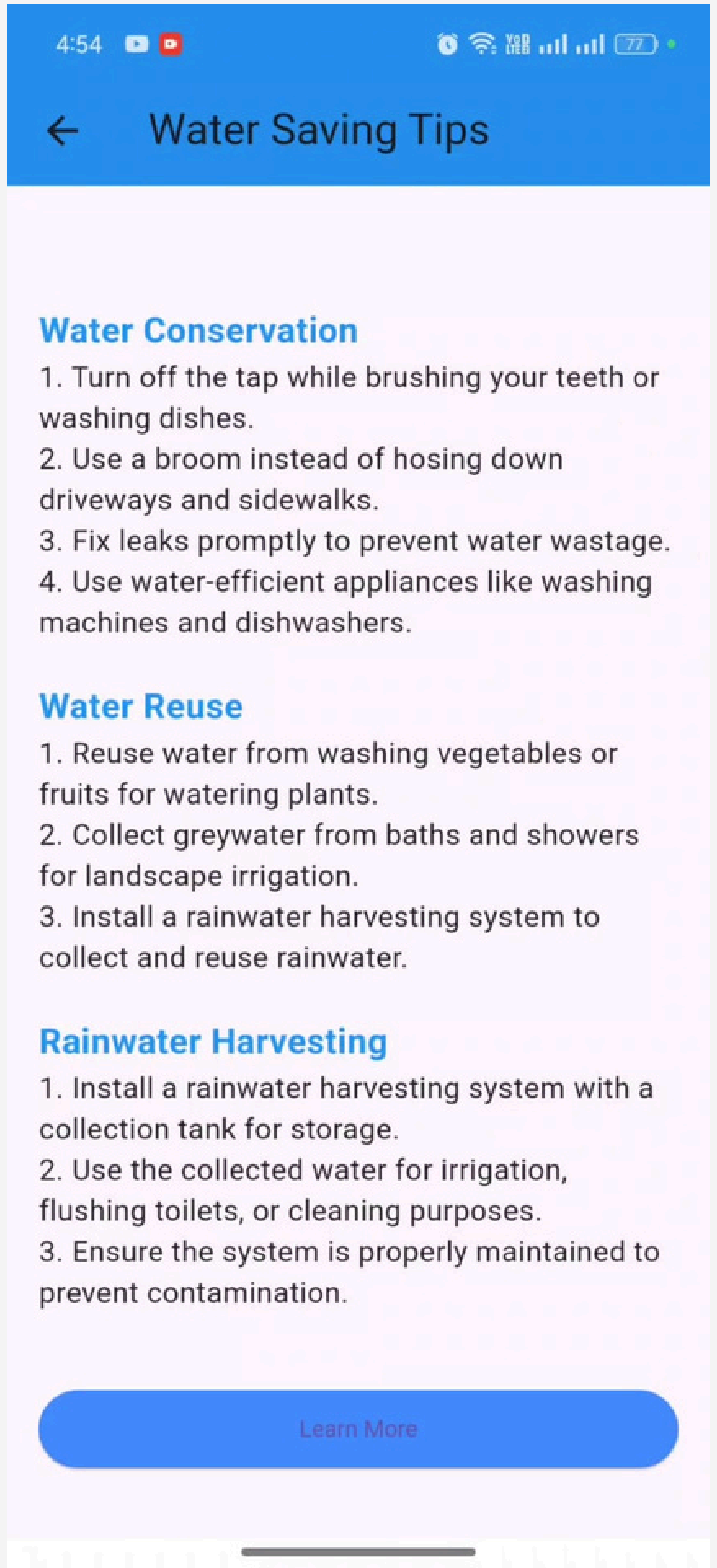


- **Water Re-use Techniques:** This page contains list of techniques farmers can use and know how to re-use waste water such as Greywater Recycling Guide, this page give information about how to use greywater recycling technique.

IRRIGATION CALCULATOR AND WATER SAVING TIPS PAGE



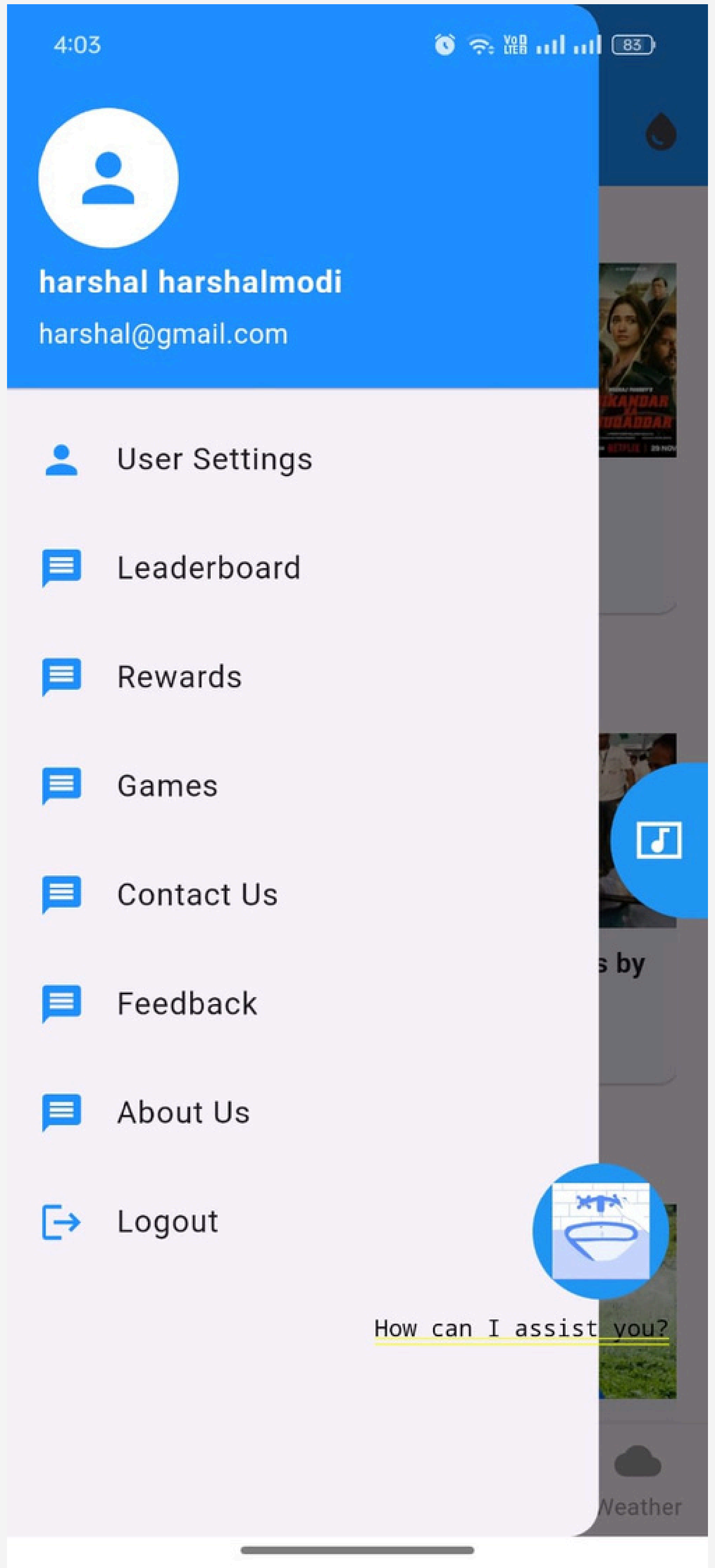


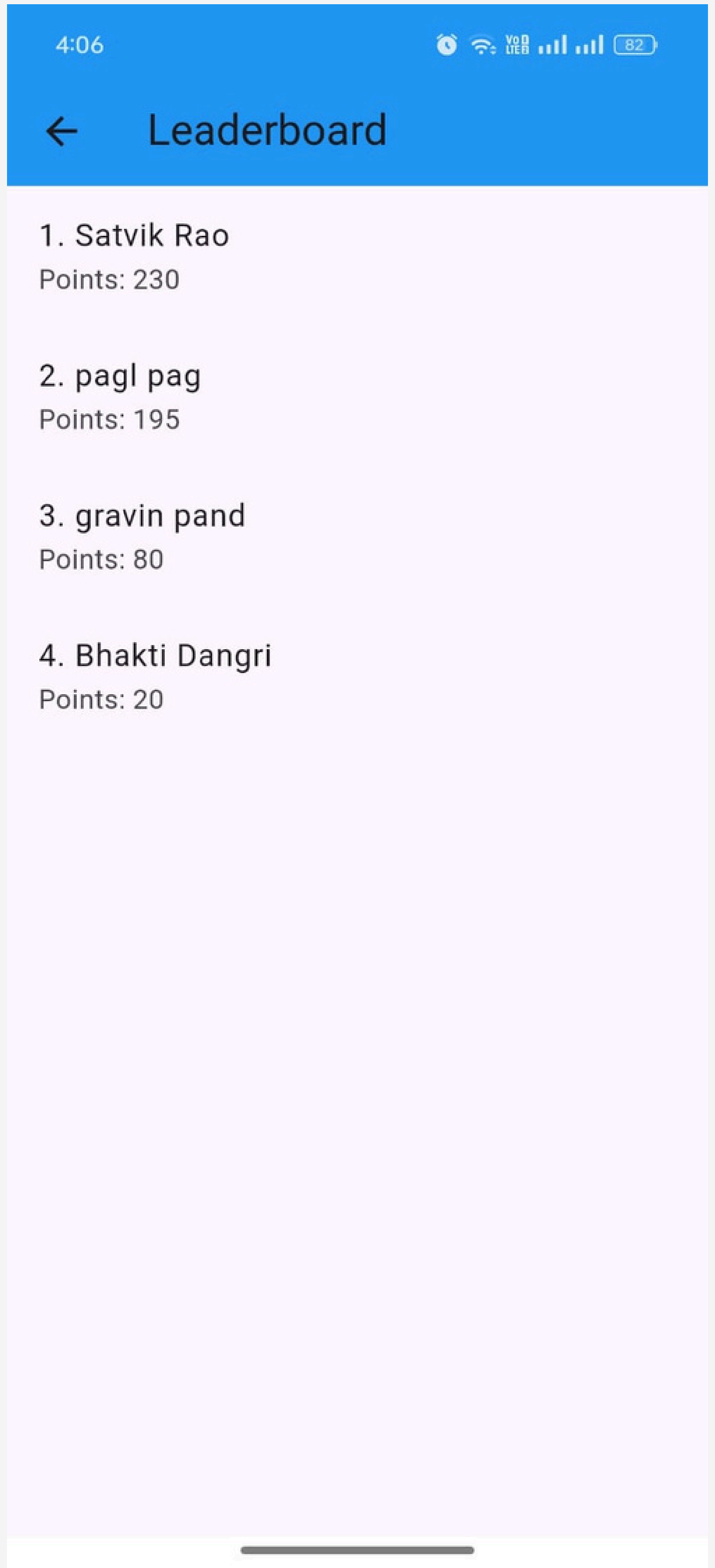


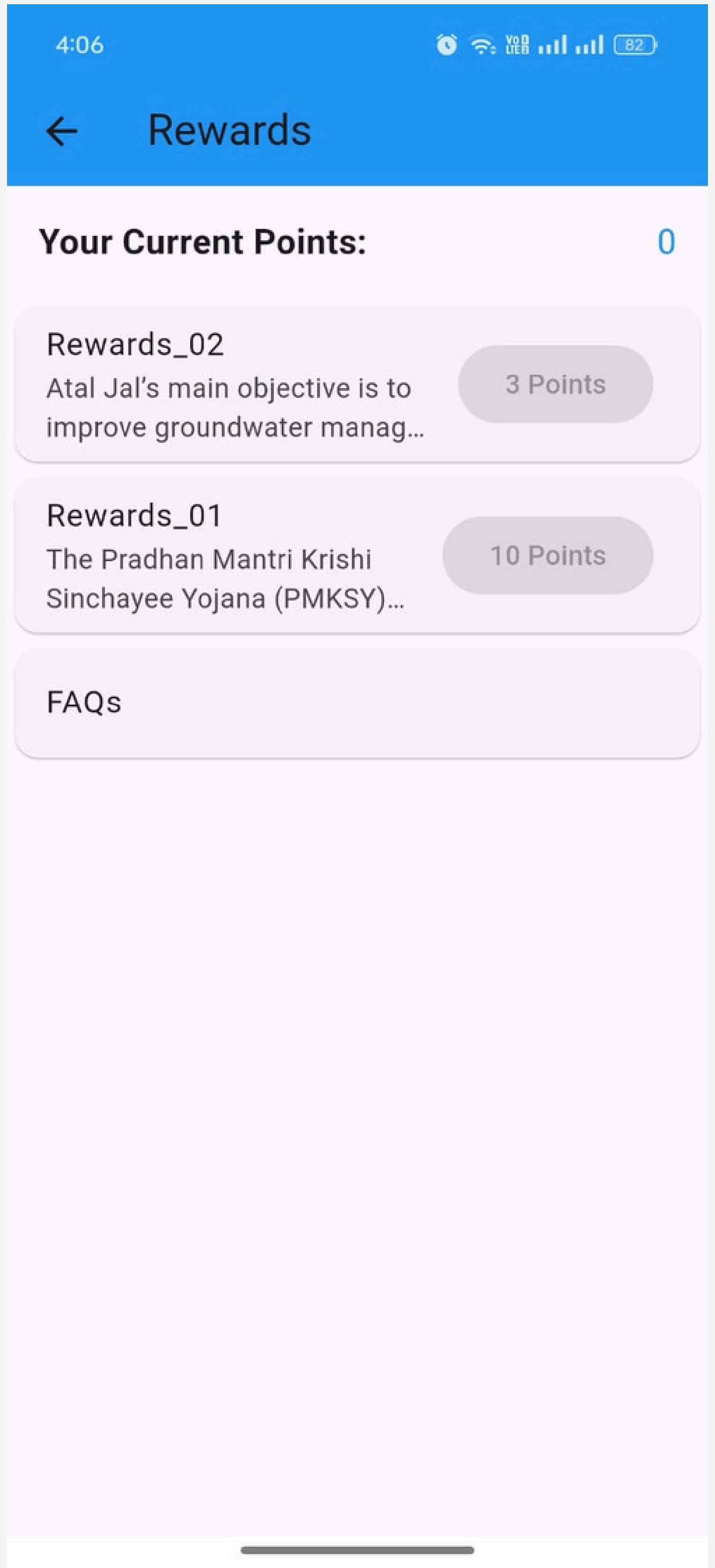
- **Irrigation Calculations Page:** Users can calculate the water requirements for their farm based on specific assumptions. Users input the farm area, and the app provides the gross water requirement and total water volume needed for the crops.

- **Water Saving Tips Page:** Users can access tips on water conservation, water reuse, and rainwater harvesting to promote sustainable water usage practices. This page provides practical and actionable tips for both home gardeners and farmers to reduce water wastage and promote sustainable water management practices.

DRAWER MENU & LEADERBOARD & REWARDS

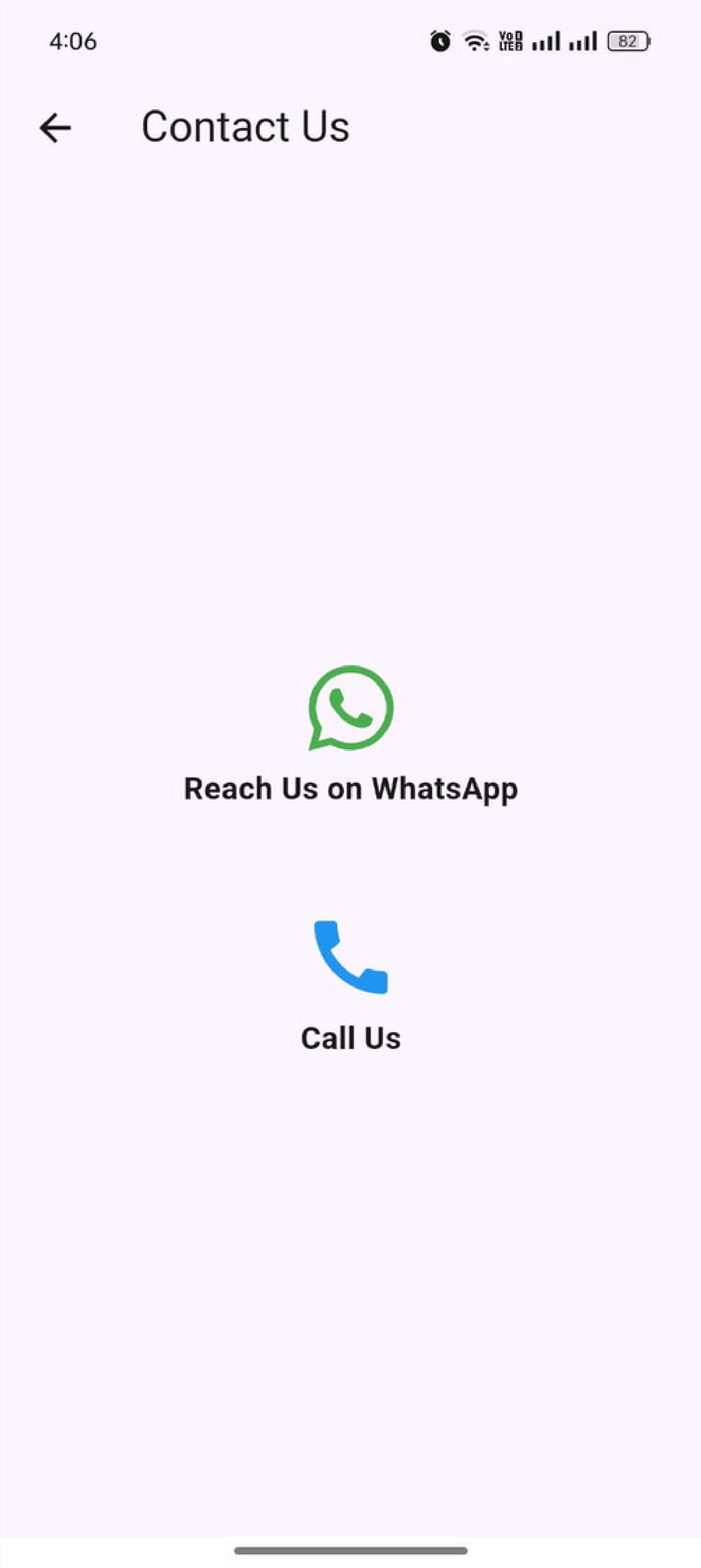


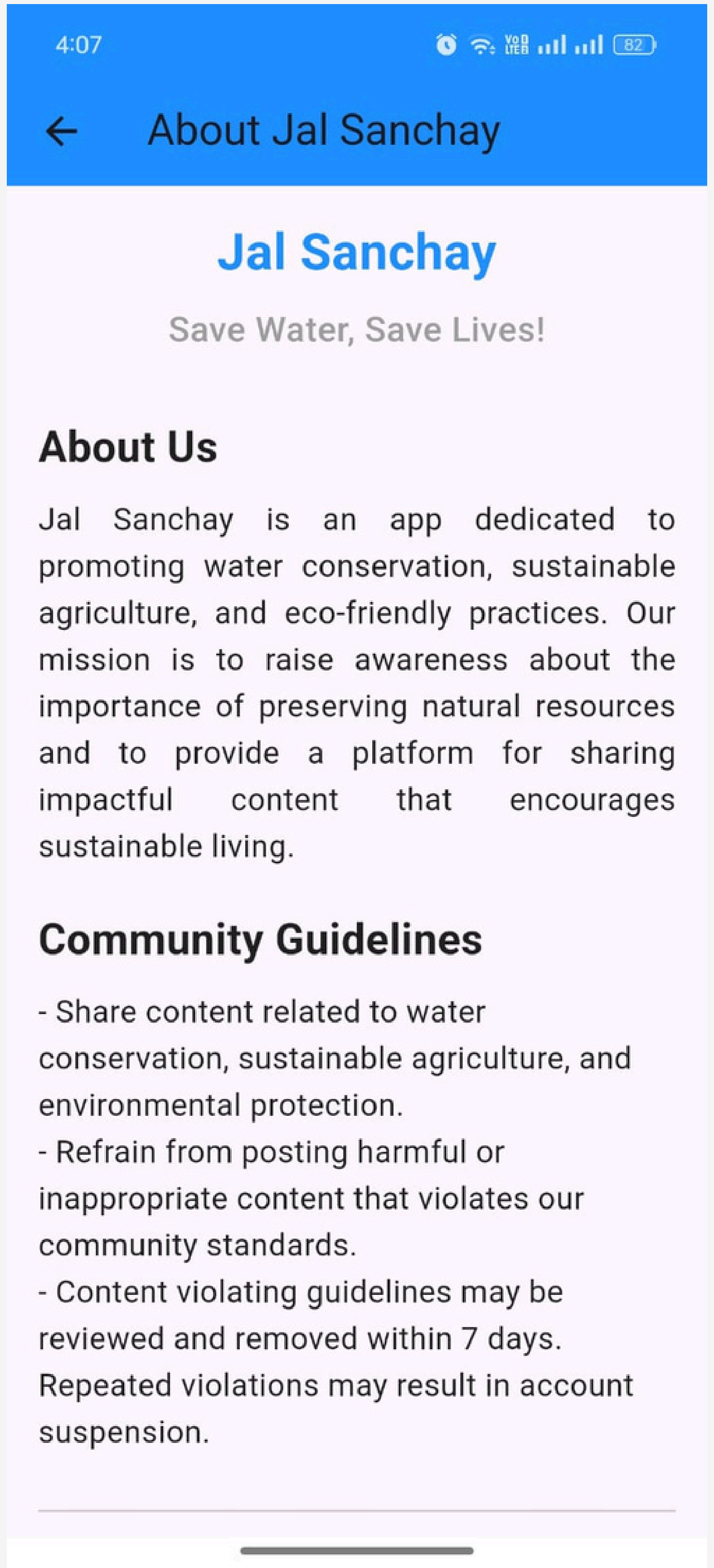


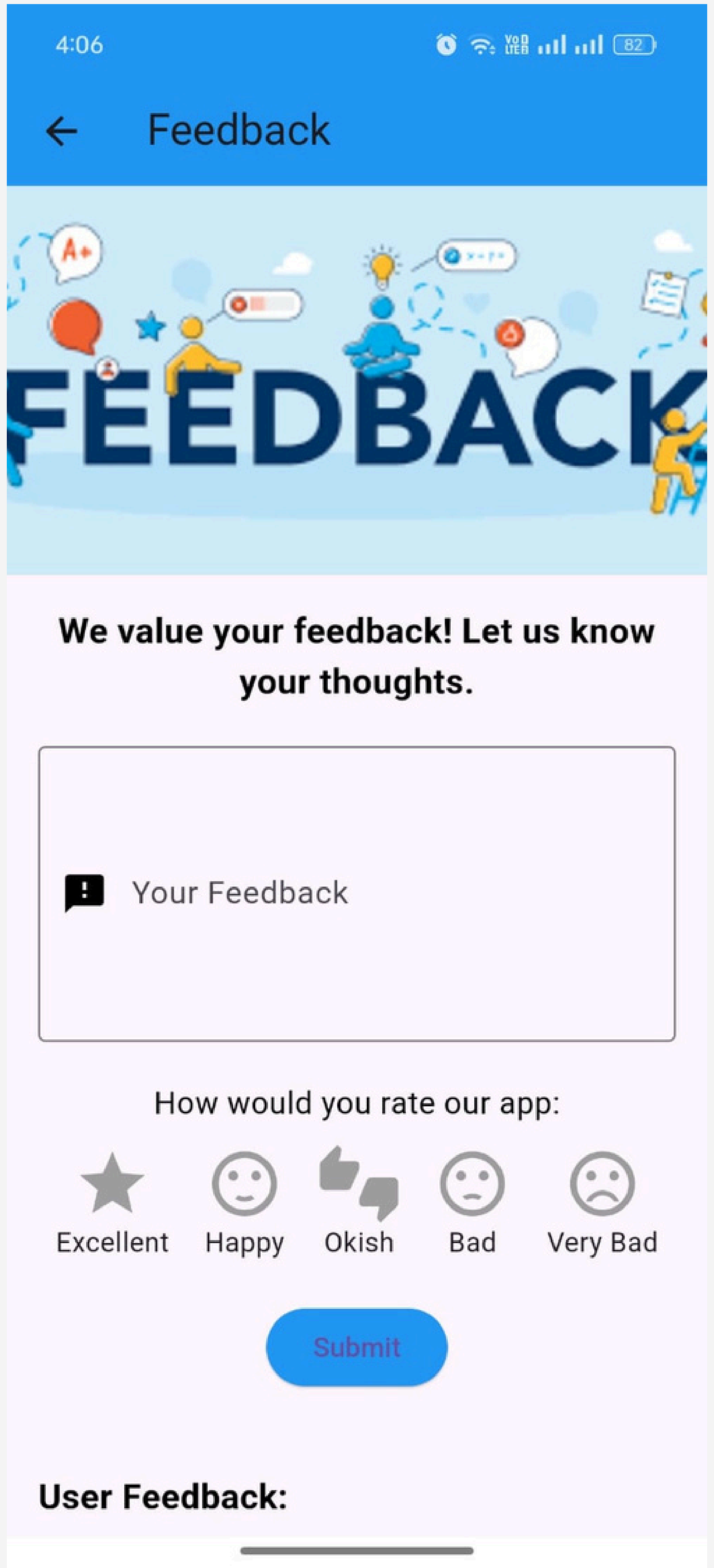


- **Drawer Menu:** The menu is on the homepage and it connects to various pages like User Settings, Games, Feedback, etc.
- **Leaderboard:** This page give overall leaderboard of all the user which have accumulated points via various actions throughout the app.
- **Rewards:** From this page we can redeem various rewards available in the app using the points we collected.

CONTACT US , ABOUT US & FEEDBACK PAGE







- **Contact Us Page:** This page enables user to contact support team using either WhatsApp chat or via call.
- **About Us:** This page gives information about the app and its community guidlines
- **Feedback:** This page is for sharing feedback, questions, or tips related to water-saving techniques and practices.